



Deep learning for new fashion product demand prediction: integrating visual similarity and demand correction in cold-start scenarios

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Abstract

Forecasting demand for new fashion items poses a persistent challenge in retail analytics, primarily due to the unavailability of historical sales data. This study presents a hybrid deep learning architecture tailored to address the cold-start problem by combining visual similarity analysis with correction techniques for censored sales data. The proposed approach utilizes multi-modal inputs, including high-resolution product images and structured metadata (e.g., category, fabric, and price), in addition to sales and inventory records that are affected by stockout-related censoring. The architecture is composed of two integrated pipelines. The first applies the Expectation–Maximization (EM) algorithm to adjust sales records, estimating true latent demand by accounting for censored entries. The second pipeline extracts deep visual features through a modified ResNet-50 architecture, followed by compression via an autoencoder to generate compact, semantically meaningful representations. These representations are clustered to identify visually similar products, enabling demand estimation through similarity-weighted averaging and localized regression within each cluster. Prediction uncertainty is captured using a bootstrap-based resampling technique. This work offers several notable contributions: the fusion of EM-based demand correction with deep visual learning, the introduction of an adaptive similarity-based forecasting strategy, and the integration of uncertainty estimation for cold-start scenarios. Empirical evaluation on a real-world multimodal fashion dataset demonstrates that the proposed model significantly outperforms several baseline approaches—including those relying solely on metadata, visual features, or global regressors—with results achieving RMSE of 10.95, MAE of 8.54, MAPE of 16.5%, Precision@10 of 73.9%, and Recall@10 of 70.7%. These outcomes underscore the effectiveness and practical value of the architecture in predicting demand for newly launched fashion products.

Keywords Fashion analytics · Deep learning · Visual similarity · EM algorithm · Product clustering

1 Introduction

Demand forecasting in the fashion retail industry represents a significant and ongoing challenge due to the dynamic nature of consumer preferences and rapid changes in product offerings. The cold-start problem, where new fashion products lack historical sales data, continues to limit the performance of conventional forecasting techniques [1]. Traditional statistical models and time series methods fail to provide accurate

demand estimates in such scenarios, especially within fast-fashion contexts characterized by short product life cycles and high variability [2].

Several machine learning and deep learning approaches have been introduced to address this issue. One line of research focuses on learning visual similarities between fashion products using deep Convolutional Neural Networks (CNNs), which extract semantic representations from product images to infer demand patterns. These methods allow the comparison of new products with previously released items to estimate demand indirectly [3]. However, these approaches often ignore the presence of censored sales data—sales records that are truncated due to stockouts or supply limitations—which distorts the actual demand distribution and leads to biased predictions [4].

Statistical correction techniques have been proposed to overcome the problem of demand censoring. Among them,

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the Expectation–Maximization (EM) algorithm has gained attention for its ability to estimate the latent demand distribution by iteratively adjusting the observed sales data. Nevertheless, these methods are typically applied in isolation and assume the availability of sufficient sales history, which does not hold for newly introduced items [5, 6].

The lack of a unified architecture that combines both visual feature-based demand inference and censored sales correction remains a notable gap in the literature. Moreover, the absence of integrated solutions that generalize well in cold-start conditions continues to hinder the reliability of forecasting models in real-world applications [7, 8].

Previous hybrid approaches have either applied censored demand correction independently or relied exclusively on visual similarity modeling [9]. The proposed framework introduces a unified integration where the EM-based correction pipeline operates in conjunction with deep visual embeddings and cluster-specific demand transfer. In this design, corrected demand values enhance the reliability of cluster formation, while the clustering structure guides the estimation of latent demand. This bidirectional interaction establishes a coherent end-to-end architecture that has not been jointly explored in prior studies [10].

The evolution of deep visual understanding has enabled more effective modeling of fashion products, particularly through the use of pre-trained convolutional networks fine-tuned for specific domain characteristics [11]. These models capture intricate visual cues, such as texture, color, and shape, which are strongly correlated with customer preferences. When combined with clustering techniques, these image embeddings can serve as a foundation for discovering latent demand structures among products. This approach is especially useful in cases where product metadata are limited or unstructured [12, 13].

Recent advances in hybrid models have highlighted the importance of combining data-driven learning with statistical corrections to improve forecasting performance under uncertainty. However, few studies have explored integrating censored demand correction with image-based feature extraction in a cohesive architecture. A comprehensive solution that accounts for both the visual characteristics of products and the statistical nature of sales data remains under-explored [14, 15].

This study presents a forecasting architecture that incorporates a censored sales correction mechanism based on the EM algorithm alongside deep visual similarity learning using a fine-tuned CNN model. The proposed method enables the assignment of new fashion products to demand clusters derived from corrected sales behavior and visual features. This combination improves the generalization of demand predictions, particularly for products without prior sales records.

The main contribution of this paper is as follows:

- A hybrid architecture is introduced that integrates censored sales correction via the Expectation–Maximization (EM) algorithm with deep image similarity learning.
- A clustering-based module is designed to assign new fashion items to groups sharing visual and behavioral similarity, enabling cold-start demand forecasting.
- A fine-tuned InceptionV3 model is employed for extracting high-level image embeddings to represent semantic attributes of fashion products.
- The proposed architecture demonstrates superior forecasting accuracy in comparison with models relying solely on either historical sales or image-based features.
- The novelty of this work lies in its integration strategy, where EM-based demand correction, deep visual embeddings, and clustering are combined into a unified end-to-end pipeline, achieving synergetic improvements beyond applying these methods in isolation.

The rest of this paper is structured as follows: Sect. 2 reviews related work on demand forecasting and image-based similarity learning. Section 3 outlines the proposed hybrid architecture, including censored sales correction and image feature clustering. Section 4 presents the experimental results and performance evaluation. Section 5 concludes the study and suggests future research directions.

2 Related work

Forecasting demand for new fashion products poses significant challenges due to the absence of historical sales data, data censoring, and the influence of non-quantitative factors such as visual appeal. To address these challenges, researchers have explored a variety of approaches, including censored demand estimation, deep learning for visual feature extraction, and hybrid models that combine structured and unstructured data [16, 17]. This section reviews key studies in these areas, highlighting their methodologies, contributions, and limitations in the context of demand forecasting for fashion products.

Steeneck et al. [18] were among the first to address censored sales data in retail environments using an Expectation–Maximization (EM) algorithm to estimate lost sales. Their work provided a foundation for modeling demand when stockouts occur, but their approach was primarily focused on fast-moving consumer goods rather than fashion items. Building on this, Hickman et al. [19] extended EM-based methods to incorporate demand learning in assortments with limited inventory, showing effectiveness in early-stage sales prediction. However, these models relied heavily on tabular sales data and lacked integration of visual information, which is critical in fashion domains.

Visual information has been effectively incorporated into demand forecasting models through the use of deep convolutional neural networks (CNNs), as demonstrated by Hung et al. [20], who introduced a visually aware recommendation architecture that learns high-level representations from product images to improve prediction accuracy. Their approach captured style preferences and improved recommendations for visually driven categories such as clothing. Similarly, Teper et al. [21] applied Siamese networks for visual similarity learning, enabling clustering of new products based on appearance. These models demonstrated the importance of visual features but did not address demand estimation directly, particularly under conditions of sales censorship.

Beyond CNN-based approaches, recurrent neural networks (RNNs) and long short-term memory networks (LSTMs) have been widely adopted in retail forecasting tasks. For instance, Bandara et al. [22] demonstrated the effectiveness of LSTMs in modeling seasonal and promotional effects across product categories. Similarly, Lin et al. [23] proposed a temporal convolutional network for multi-horizon forecasting, achieving state-of-the-art performance on retail datasets. More recently, Zhou et al. [24] applied Transformer-based architectures to capture long-range temporal dependencies in sales data, showing improved accuracy compared to RNN-based models. However, these approaches primarily rely on historical time series and are less effective in cold-start scenarios where no sales history is available.

Several studies have explored hybrid models that combine visual features with sales data. Mousavizadegan et al. [25] introduced a Bayesian censored regression model that integrates image-derived features into demand forecasting for online fashion retail. Their results showed that visual embeddings can improve predictions when sales data are sparse. However, their method required large labeled datasets and did not include any clustering mechanism for demand transfer across similar products.

In the context of product clustering, Lan et al. [26] utilized unsupervised deep clustering to group fashion items based on visual similarity, allowing for group-level demand estimation. While this approach improved cold-start prediction, it struggled to incorporate censored sales correction. More recently, Kaur et al. [27] applied deep metric learning to learn image similarity in fashion retail, employing triplet loss to capture fine-grained differences in style. Despite their success in feature learning, their architecture lacked a probabilistic correction mechanism for stockout-affected sales.

Beyond clustering-based methods, transfer learning has been explored to mitigate the cold-start problem. Celik et al. [28] utilized meta-learning to adapt forecasting models to new product launches with minimal data. Kassem et al. [29] proposed a transfer learning framework that leverages knowledge from related product categories to improve prediction for unseen items. More recently, Baier et al. [30] investigated

the use of large language models to incorporate unstructured textual descriptions and user-generated content into demand forecasting, highlighting the potential of multimodal integration. While these methods improve adaptability, they often require extensive auxiliary data and computational resources.

Another line of research focused on forecasting models that explicitly handle demand censoring. Yavuz et al. [31] proposed a deep demand learning model that adjusts for inventory constraints using reinforcement learning. Although their method achieved high accuracy in constrained environments, it required extensive tuning and did not generalize well to new products without prior sales.

Beyond EM-based and reinforcement learning approaches, Bayesian censored regression models have been introduced to estimate lost sales under stockout conditions by Trapero et al. [32]. In parallel, Li et al. [33] proposed survival analysis techniques have been applied to model the probability of product unavailability over time, offering robustness in sparse demand environments. Despite their effectiveness, these methods are computationally intensive and may be difficult to scale in fast-moving retail contexts.

Despite the variety of techniques employed across these studies, a major limitation persists: Few works jointly address the issue of censored demand while leveraging visual features to forecast demand for new or cold-start products. Notably, all the studies referenced in this comparison are from 2025, reflecting the most recent advancements and ongoing interest in this domain. Table 1 summarizes key contributions and limitations of these state-of-the-art approaches. Building upon this recent body of work, the proposed architecture integrates EM-based censored demand correction with deep visual similarity learning, providing a comprehensive and robust solution for demand forecasting in cold-start fashion scenarios.

3 Proposed methodology

This section presents a hybrid architecture for demand forecasting of new fashion products without historical sales, as illustrated in Fig. 1. The proposed methodology combines two key components: censored sales correction using the Expectation–Maximization (EM) algorithm and visual similarity modeling using deep convolutional neural networks (CNNs). The architecture begins by exploring a multimodal fashion dataset that includes product images and associated metadata. Sales records for existing products are first corrected using the EM algorithm to account for stockouts and reveal latent demand patterns. In parallel, visual features are extracted from product images using a pretrained CNN, enabling the grouping of visually similar items. By matching new products to these clusters, the model transfers corrected demand signals from similar historical products

Table 1 Summary of State-of-the-Art Demand Forecasting Studies in the Fashion Domain in 2025

Article	Technique	Problem Addressed	Short Description	Accuracy Level	Key Limitation	Advancement in Current Work
Sousa et al. [34]	EM Algorithm	Censored Demand Estimation	Estimates lost sales via iterative EM	Medium	No visual product representation	Combines EM with deep image embedding
Gautam et al. [35]	Bayesian Demand Estimation		Uses Bayesian inference to estimate censored demand	Medium	No visual product differentiation	Applies statistical estimation within visual clusters
Khan et al. [36]	CNN + RNN		Jointly models images and temporal features	Medium	Ignores lost demand from stockouts	Integrates demand correction into visual-temporal modeling
Sabitha et al. [37]	Triplet Loss + CNN	Visual Similarity Matching for Cold-Start	Learns visual similarity for new products	Medium	No consideration of availability/sales constraints	Adds sales correction to similarity pipeline
Lu et al. [38]	Variational Autoencoder (VAE)		Learns latent style embeddings from images	High	Ignores stock-induced bias	Combines visual modeling with EM correction
Mrinali et al. [39]	Vision Transformer (ViT)		Uses ViT to extract image features	High	High complexity; lacks demand correction	Replaces ViT with lightweight CNN + correction module
Wang et al. [40]	Clustering + ARIMA	Forecasting Without Historical Sales	Applies clustering then ARIMA for forecasting	Low	Ineffective with cold-start items	Uses deep image clustering without historical sales
Liu et al. [41]	GNN for Product Graphs		Learns demand using co-purchase and visual links	High	Depends on user–item interactions	Functions independently of behavioral data
Li et al. [42]	Deep Autoencoder	Dimensionality Reduction + Forecasting	Compresses features before regression	Medium	Weak with incomplete/censored data	Joint autoencoding, imputation, and similarity estimation
Hwang et al. [43]	Random Forest + Ensemble		Applies tree ensembles to structured data	Medium	Poor generalization to unseen items	Enables image-driven generalization for new SKUs

to estimate future demand. This integration of statistical correction and deep visual representation offers a robust approach for forecasting demand in cold-start scenarios. In addition to visual embeddings, structured product attributes such as brand, price, fabric type, and seasonality are incorporated into the representation space. These attributes are concatenated with the visual features to construct a multimodal embedding, ensuring that the forecasting process

reflects both appearance-driven and non-visual consumer preferences. Each component of the architecture is detailed in the subsequent section.

3.1 Dataset selection

This study employs the VISUELLE dataset, a large-scale fashion dataset specifically designed to support research in

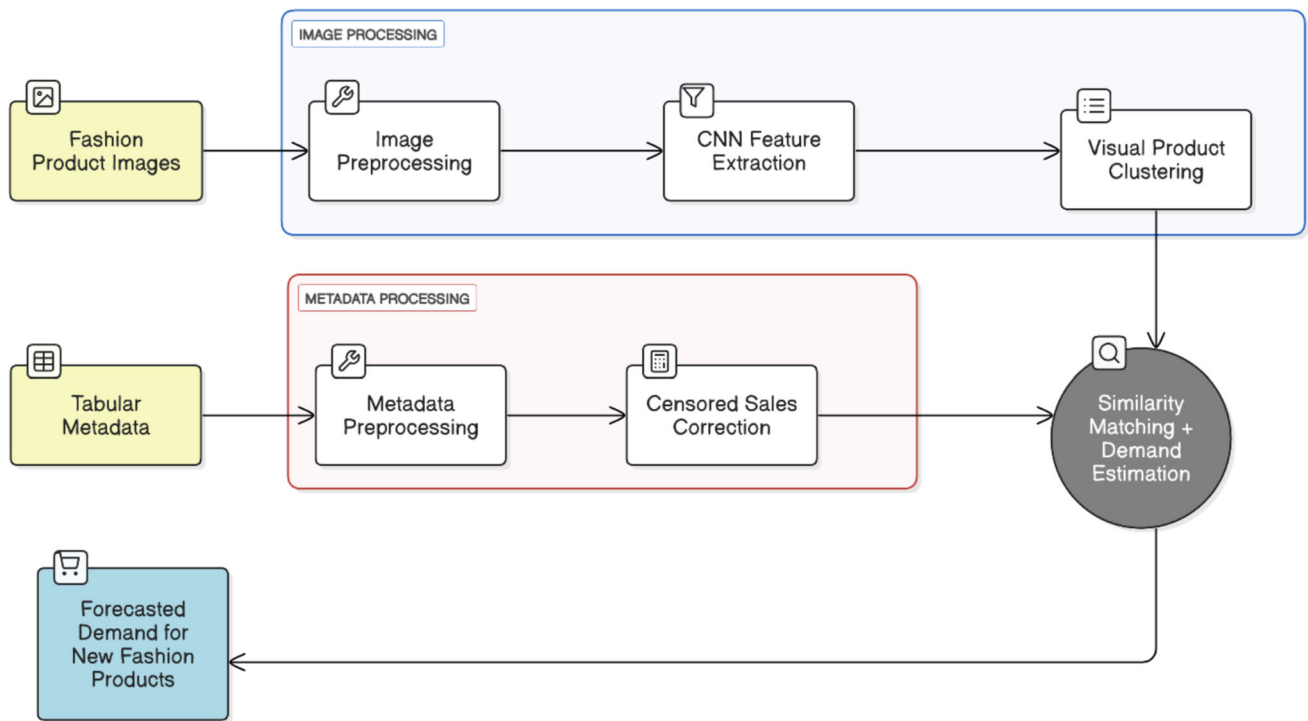


Fig. 1 Abstraction of the Proposed Methodology

demand forecasting for newly released clothing items. The dataset consists of 44,000 fashion products covering 144 distinct categories and is curated from real-world fashion e-commerce platforms. Each entry is paired with a high-resolution image and rich metadata, enabling multimodal learning strategies that integrate both visual and structured features. The primary objective of using this dataset is to facilitate robust demand forecasting models capable of making predictions for new products without relying on historical sales records [44].

The dataset provides a comprehensive set of features including product image, category, dominant color, fabric type, seasonality, launch date, brand, and retail price. Furthermore, it incorporates time series data such as sales trends and stock availability post-launch. To enhance the forecasting models with contextual demand signals, the dataset also includes auxiliary information like Google Trends data for categories, colors, and fabrics. This integration of image data with temporal and semantic signals creates a robust foundation for capturing visual appeal, consumer interest, and market behavior. For all experiments, the forecasting horizon was fixed to four weeks ahead of the product launch. This time window aligns with standard retail planning cycles and allows for meaningful comparison with baseline methods. An overview of the dataset features is presented in Table 2, which summarizes the available attributes along with their respective types and descriptions.

A representative sample of product images is shown in Fig. 2, illustrating the diversity of styles, colors, and garment types present in the VISUELLE dataset. These examples emphasize the dataset's suitability for modeling visual similarity and clustering in fashion-related applications. This multimodal richness makes VISUELLE particularly suitable for hybrid demand forecasting strategies that leverage both structured attributes and deep image features.

3.2 Data preprocessing

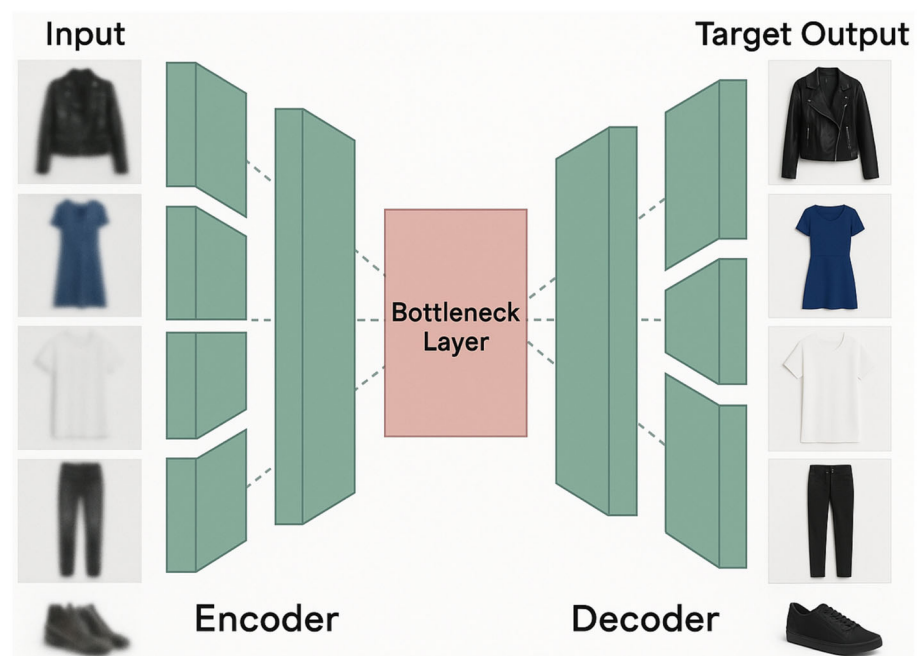
This study employs a deep learning-based preprocessing pipeline that enhances the quality and consistency of visual input data prior to feature extraction. This pipeline aims to improve the structural clarity and semantic fidelity of the images, ensuring that downstream models receive inputs that are both visually informative and statistically stable. Given the variability in photographic quality, background environments, and lighting conditions within fashion datasets, a robust preprocessing strategy is essential for effective representation learning.

The first stage of the pipeline utilizes a Convolutional Denoising Autoencoder (CDA) to reconstruct clean visual representations from noisy input images. The architecture of the CDA consists of a symmetrical encoder–decoder structure. The encoder is composed of four convolutional layers, each followed by batch normalization and ReLU activation. The number of filters increases progressively from 32 to 128,



Fig. 2 Sample product images from the VISUELLE dataset illustrating category and style diversity

Fig. 3 Example of autoencoder-based image reconstruction using product images from the VISUELLE dataset



and a stride of 2 is used to reduce spatial dimensions while capturing hierarchical features. The latent space serves as a compact abstraction of the original image, preserving key garment characteristics such as shape, texture, and style. The decoder mirrors the encoder with four transposed convolutional layers, which gradually restore the spatial resolution. A final sigmoid activation is used to generate output images with normalized pixel values between 0 and 1. The CDA is trained in an unsupervised fashion using a large sample of the dataset and is optimized to reconstruct denoised versions that retain the stylistic essence of the original garments. A visual

representation of the CDA architecture and its transformation process is illustrated in Fig. 3, which demonstrates the encoder–decoder flow from input to reconstructed output.

Following denoising, a perceptual normalization step is applied to further align images in a high-level semantic space. Rather than standardizing based on raw pixel values, this process leverages the feature representations extracted from intermediate layers of a pre-trained VGG16 network. Each image is passed through the network to obtain multi-level activations, which are then normalized relative to reference statistics computed from the dataset. This technique reduces distributional shifts in the visual features and enhances model stability during training. Formally, given an image x and its

Table 2 Overview of the VISUELLE Dataset Features

Feature	Description	Data Type
Product Image	High-resolution image of the fashion item	Image
Category	Product type (e.g., dress, top, pants)	Categorical
Color	Dominant product color	Categorical
Fabric Type	Main fabric material (e.g., cotton, silk)	Categorical
Brand	Product's brand	Categorical
Season	Season of product release (e.g., Spring 2018)	Categorical
Launch Date	Date the product was first introduced	Date
Price	Retail price in Euros	Numeric
Sales Time Series	Daily/weekly sales post-launch	Numeric (time series)
Stock Availability	Daily stock level for each product	Binary/Numeric
Google Trends—Category	Search trend index for the product category	Numeric
Google Trends—Color	Search trend index for the dominant color	Numeric
Google Trends—Fabric	Search trend index for the main fabric	Numeric

perceptual representation $\phi_l(x)$ at layer l , the normalization process is defined as Eq. (1):

$$(x)^{norm} \phi_l = \frac{\phi_l(x) - \mu}{\sigma_l} \quad (1)$$

where μ and σ_l represent the mean and standard deviation of activations at layer l across the dataset. The perceptually normalized image features are then projected into a common latent space, ensuring consistent stylistic encoding. An optional whitening transformation may be applied to further decorrelate feature channels defined as Eq. (2):

$$\phi_l^{white}(x) = \phi_l^{norm}(x) \cdot W_l \quad (2)$$

where W_l is the whitening matrix derived from the covariance structure of $\phi_l(x)$. This stage prepares the dataset for robust feature extraction and high-fidelity similarity learning.

The VGG16 model was employed in a feature extraction mode without end-to-end fine-tuning, as its convolutional layers are well-suited for capturing perceptual features such

as edges, textures, and color distributions. Specifically, activations from the conv3_3 and conv4_3 layers were used for normalization, as these layers balance low-level detail with mid-level stylistic abstractions relevant to fashion imagery.

3.3 Feature extraction and clustering

In order to model visual similarity among fashion products and prepare them for subsequent demand forecasting, a multi-stage process is employed comprising feature extraction, dimensionality reduction, and unsupervised clustering. This pipeline captures the intrinsic visual semantics of each product image and enables the organization of items into coherent stylistic groups.

3.3.1 Deep feature extraction using modified ResNet50

Convolutional Neural Networks (CNNs) are widely adopted for extracting hierarchical representations from visual data, particularly in domains that require fine-grained semantic understanding such as fashion analysis. In this study, we employ a modified version of the ResNet50 architecture, a deep residual CNN originally designed to alleviate vanishing gradient problems in very deep networks through the use of identity-based skip connections. ResNet50 is composed of 50 layers, including convolutional, batch normalization, ReLU, and identity shortcut connections, making it suitable for capturing multi-level abstractions of visual patterns.

The modified ResNet50 architecture is adapted for fashion product feature extraction by removing its original classification head and introducing a lightweight embedding layer tailored for deep similarity modeling. The backbone, pretrained on ImageNet, remains frozen during the initial training phase to retain general-purpose visual representations suitable for downstream clustering and retrieval tasks. After the last convolutional block (conv5_x), a global average pooling layer is applied to reduce spatial dimensions while retaining channel-wise activations. This is followed by a fully connected (dense) layer with 512 neurons, which projects the features into a lower-dimensional latent space. A dropout layer with a rate of 0.5 is inserted before the dense layer to mitigate overfitting. Finally, a L2 normalization layer ensures that the output embeddings lie on a unit hypersphere, making them suitable for similarity-based learning.

In addition to the ResNet-50 embeddings, intermediate activations from the VGG16 backbone specifically block4_conv3 and block5_conv3 were extracted to capture complementary perceptual features such as texture and fine-grained appearance cues. The 512-dimensional embedding from ResNet-50 was concatenated with a 512-dimensional vector derived from VGG16, resulting in a joint 1024-dimensional representation. This multimodal visual

descriptor was then passed to the autoencoder for nonlinear dimensionality reduction.

The output from this modified architecture is a 512-dimensional embedding vector for each image, encoding high-level visual cues such as garment structure, texture, silhouette, and stylistic components. These embeddings are used in subsequent stages for clustering and similarity matching. During training, contrastive learning is optionally applied using a SimCLR-inspired objective, where positive pairs are generated through heavy data augmentation, and a contrastive loss function is used to maximize agreement between positive pairs while minimizing it between negative ones.

In this work, positive pairs are generated by applying multiple stochastic augmentations such as cropping, flipping, and color jittering to the same image instance, ensuring invariance to these transformations. Negative pairs are sampled from different product categories, with priority given to visually dissimilar items such as footwear versus outerwear, to provide stronger contrastive signals. This sampling strategy enforces that the encoder learns to cluster semantically similar items while separating unrelated products in the embedding space.

This training strategy enhances the model's ability to distinguish fine-grained variations between fashion products, which is crucial for accurate demand estimation. The classification head is replaced with a global average pooling layer, dropout, and a 512-dimensional dense projection layer with L2 normalization to produce compact image embeddings. A schematic overview of the modified ResNet50 architecture is shown in Fig. 4, where the embedding head is delineated from the frozen convolutional backbone and the contrastive learning branch is explicitly illustrated. The figure highlights how positive pairs are generated through heavy data augmentation of the same product image, while negative pairs are sampled from different categories, enforcing discriminative learning in the embedding space. This representation forms the basis for clustering and cold-start demand prediction.

The ResNet-50 backbone was initially frozen to retain general-purpose representations and subsequently fine-tuned on the VISUELLE dataset with domain-specific augmentations such as color jittering, fabric texture enhancement, and style-preserving transformations. Fine-tuning was applied to the last two residual blocks conv4_x and conv5_x, allowing the model to adapt high-level abstractions to fashion-specific features such as garment silhouette, fabric texture, and stylistic variations. This process significantly improved the discriminative capacity of the embeddings for downstream clustering and similarity-based forecasting.

3.3.2 Dimensionality reduction using autoencoder

This study employs a deep autoencoder architecture to compress high-dimensional visual feature vectors extracted from fashion product images, offering a nonlinear dimensionality reduction technique. In contrast to conventional methods such as PCA or t-SNE, which rely on linear transformations or local neighborhood structures, the autoencoder learns complex task-specific latent representations through end-to-end training, enabling more expressive and semantically rich embeddings.

The employed autoencoder follows a symmetric encoder-decoder structure, specifically designed to process 1024-dimensional feature vectors obtained from a modified ResNet-50 backbone. The encoder begins with a fully connected (FC) layer that reduces the input dimension from 1024 to 512, followed by a second FC layer that maps the features to 128 dimensions. Each of these layers is followed by a ReLU activation function and batch normalization to ensure both nonlinearity and training stability. At the core of the model lies a latent embedding layer with 64 dimensions, serving as the compressed representation of the original feature vector. This bottleneck captures the most salient visual characteristics while discarding redundant or non-informative components.

The decoder mirrors the encoder's structure in reverse. It first expands the 64-dimensional latent vector to 128 dimensions through an FC layer with ReLU and batch normalization, followed by a second expansion to 512 dimensions using the same activation and normalization scheme. The final reconstruction layer restores the feature vector to its original 1024-dimensional form, ensuring that the output is aligned with the input space. The network is trained in an unsupervised manner using a mean squared error (MSE) loss function, which minimizes the difference between the input and the reconstructed output. This process forces the model to learn efficient, semantically meaningful encodings in the latent space.

The decision to employ a separate autoencoder, rather than extending the convolutional backbone, was motivated by the need for flexibility and generalization. The autoencoder operates on heterogeneous feature concatenations, such as embeddings derived from both ResNet and VGG, without requiring architectural modifications to the pretrained networks. This modularity ensures that the backbone models preserve their pretrained stability, while the dimensionality reduction is handled independently in a way that is less prone to overfitting, particularly given the relatively limited size of the VISUELLE dataset. In this manner, the autoencoder provides a compact yet semantically meaningful representation that is specifically tailored to the downstream forecasting task, while maintaining the robustness of the visual feature extractors.

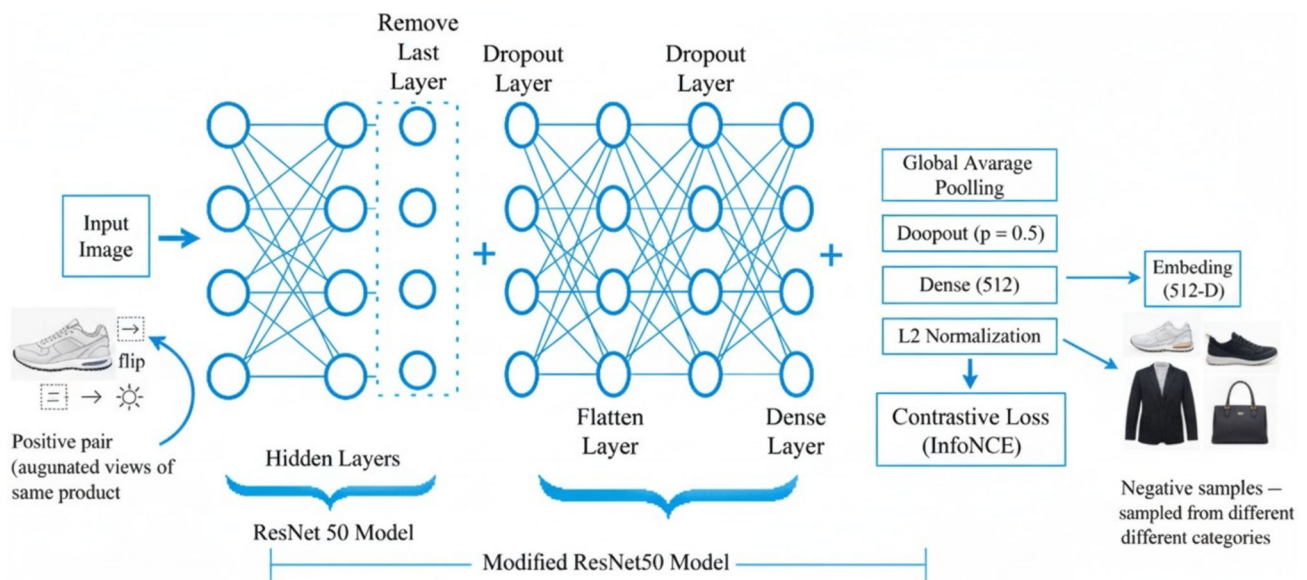


Fig. 4 Modified ResNet-50 encoder with embedding head and contrastive learning objective. Positive pairs are generated via augmented views of the same product; negative pairs are sampled from different categories (e.g., footwear, outerwear, accessories)

Although convolutional denoising autoencoders are commonly used for raw image reconstruction, the present study operates directly on 1024-dimensional feature vectors extracted by ResNet-50. For this reason, a fully connected autoencoder architecture was selected, as it is better suited for vectorized inputs. Variational autoencoders (VAEs) were also considered; however, preliminary experiments on the VISUELLE dataset showed that the stochastic sampling in VAEs produced less stable latent embeddings and weaker preservation of fine-grained stylistic cues. The deterministic autoencoder was therefore adopted to ensure more consistent and semantically robust feature compression.

In contrast to traditional linear reduction methods, the non-linear transformation capabilities of the autoencoder allow it to better capture complex structures and manifold variations inherent in fashion imagery. These learned embeddings retain essential garment-specific visual cues, making them particularly suitable for downstream tasks such as product clustering, similarity retrieval, and cold-start demand forecasting. An illustration of the implemented autoencoder architecture is presented in Fig. 5, where the encoder gradually compresses the input feature vector through successive dimensionality reductions, culminating in a 64-dimensional latent embedding. The decoder then reconstructs the original representation, enabling the model to retain key semantic features while operating in a significantly reduced space. This design aligns with the objective of efficient feature compression and semantic preservation in high-variance visual domains such as fashion.

3.3.3 Product clustering using K-means on autoencoded embeddings

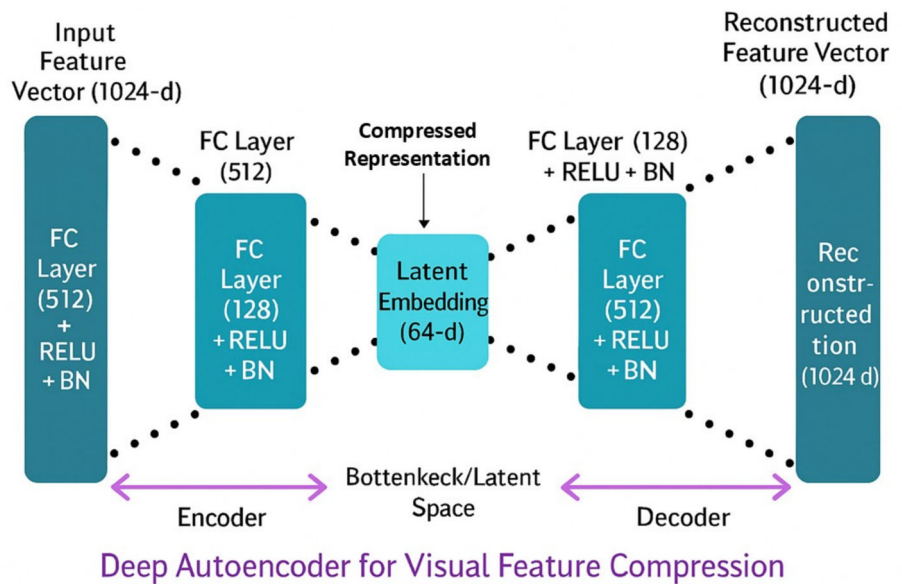
This study applies an unsupervised clustering approach to the compressed image representations generated by the autoencoder in order to organize fashion products into perceptually and behaviorally consistent groups. The learned embeddings encode high-level semantic features such as texture, structure, and style, allowing the model to group visually similar items within a shared latent space, thereby facilitating downstream tasks like similarity-based demand forecasting.

The dimensionality-reduced vectors from the bottleneck layer of the autoencoder are first normalized using L2 normalization to ensure uniformity in magnitude. These feature vectors are then passed into the K-Means clustering algorithm, which partitions the latent space into k distinct clusters. The number of clusters is selected empirically using validation metrics such as the silhouette score, ensuring the resulting groups exhibit strong intra-cluster cohesion and inter-cluster separation.

Each cluster formed through this process reflects a group of visually and stylistically related products. For new, unseen fashion items, the model extracts their embeddings using the same trained autoencoder, normalizes them, and assigns them to the nearest cluster based on cosine similarity. This enables cold-start items—those without historical sales records—to inherit behavioral patterns from similar products, forming the basis for demand estimation in subsequent stages.

Algorithm 1 summarizes the visual clustering stage of the proposed pipeline. The procedure begins with the compressed embeddings produced by the autoencoder, which are

Fig. 5 Architecture of the Deep Autoencoder Used for Nonlinear Dimensionality Reduction of Fashion Image Features



L2-normalized to ensure consistent scale across samples. K-Means is applied to the normalized embeddings to partition the latent space into k clusters, where the optimal value of k is selected empirically using validation metrics such as the Silhouette score. Each historical product is assigned to its nearest cluster centroid. For a new product, the embedding is computed using the same encoder, normalized, and mapped to the closest centroid using cosine similarity. The resulting cluster assignment enables the transfer of behavioral and demand signals from historically observed items to cold-start products and serves as the primary input for subsequent demand estimation modules. The clustering pipeline is illustrated in Fig. 6

3.3.4 Similarity matching for cold-start forecasting

In the proposed architecture, similarity matching plays a central role in addressing the cold-start problem by enabling demand inference for new fashion items that lack historical sales data. After visual embeddings are extracted for each product using the modified ResNet-50 backbone and compressed via the autoencoder, the resulting 64-dimensional vectors are L2-normalized and used for similarity-based retrieval. Specifically, for each new product, the cosine similarity is computed against all items in the training set (historical catalog), and the top-K most similar products are identified.

The similarity score between a new product embedding v_{new} and a historical product v_i is computed using the cosine similarity metric as expressed in Eq. (3):

$$S_i = \frac{v \cdot v_{i_{new}}}{\|v\| \cdot \|v_{i_{new}}\|} \quad (3)$$

These similarity scores are then transformed into weights using a softmax function as expressed in Eq. (4):

$$w_i = \frac{\exp(\alpha S_i)}{\sum_{j=1}^K \exp(\alpha S_j)} \quad (4)$$

where α is a sharpness parameter controlling the influence of the most similar items. The predicted demand is computed as the weighted sum of the corrected demand values of these top-K neighbors. This similarity matching mechanism is efficient and scalable, particularly when implemented using FAISS-based vector indexing, allowing sub-linear retrieval even in large-scale datasets.

Although this approach is effective and lightweight, alternative similarity learning techniques have been explored in the literature. For instance, Siamese and Triplet Networks learn to directly minimize the distance between similar item pairs and maximize it for dissimilar ones using contrastive or triplet losses. These methods offer more refined embeddings, particularly for fine-grained visual distinctions, but they require extensive pair or triplet mining and longer training time.

Other advanced techniques, such as Proxy-NCA or Proxy-Anchor losses, aim to improve training efficiency by representing each class with a proxy vector. These methods generate better distributed embeddings, especially when category labels are available, but their applicability becomes limited in cold-start settings where labeled data for new items are scarce.

Moreover, multimodal embedding models like CLIP align visual and textual representations in a shared latent space and have shown promising results in fashion retrieval tasks.

VISUAL CLUSTERING PIPELINE

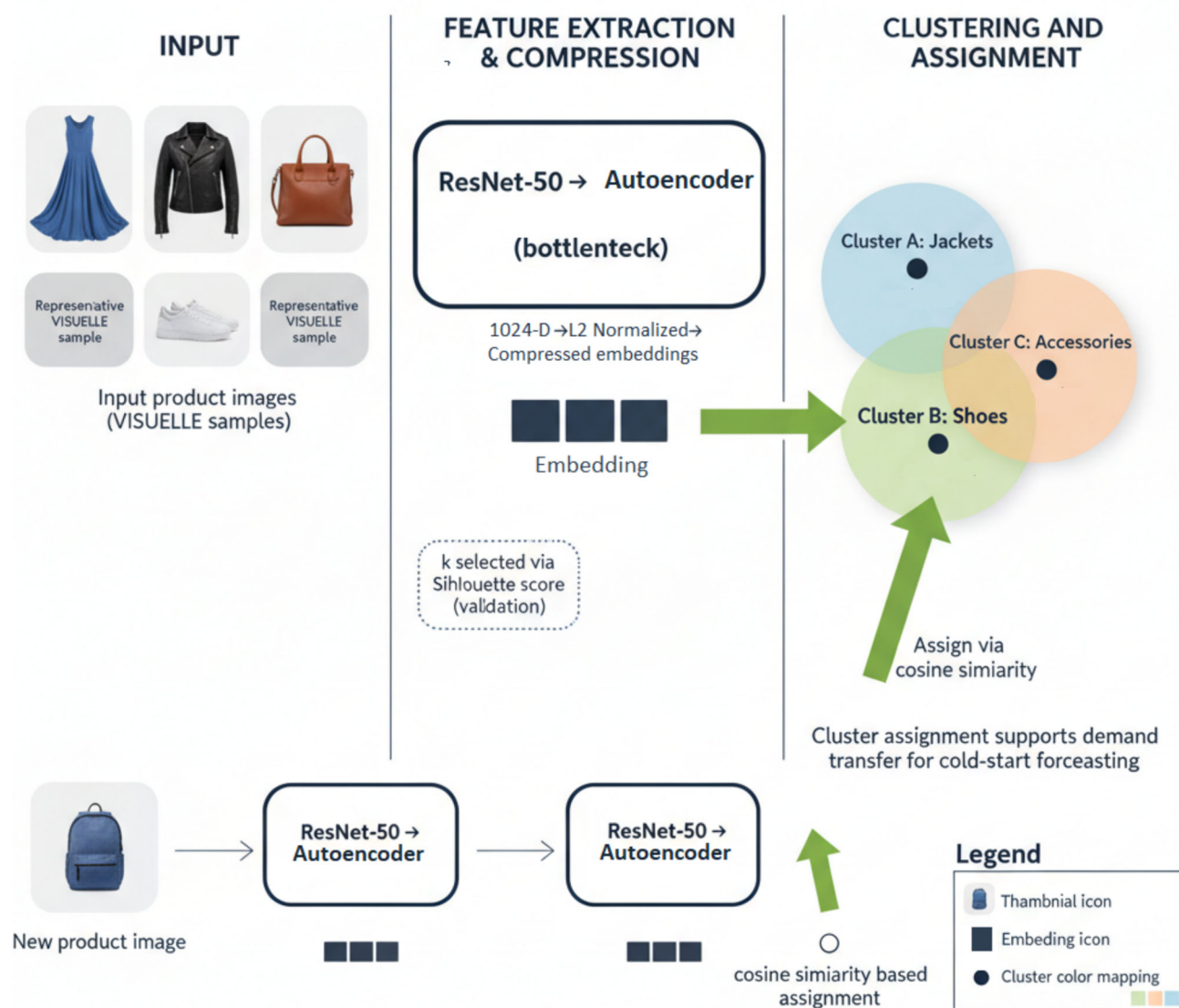


Fig. 6 Algorithm 1 (Conceptual Workflow)—Visual Clustering Using Autoencoded Embeddings for Fashion Products

However, these models require textual metadata and domain-specific fine-tuning to effectively capture fashion-related nuances, which may not always be available or reliable in structured fashion datasets like VISUELLE.

In this study, we opted for a cosine-based similarity matching approach due to its simplicity, compatibility with the rest of the pipeline, and its low inference latency. While alternative methods offer potential performance gains, they also introduce additional complexity and computational overhead. Given the cold-start nature of our task and the availability of high-quality visual embeddings, the selected similarity

matching method offers a practical trade-off between accuracy, efficiency, and integration ease.

3.4 Censored sales correction using the EM algorithm

Historical sales data in retail environments often suffer from censoring due to stockouts, where actual consumer demand exceeds the available inventory. This mismatch results in biased demand signals, particularly problematic

when building forecasting models for new products. To mitigate this issue, a censored demand correction methodology is employed to estimate the true underlying demand by reconstructing what would have been sold in the absence of inventory constraints.

The proposed methodology adopts a probabilistic architecture based on the Expectation–Maximization (EM) algorithm. The aim is to correct historical sales data by inferring latent (unobserved) demand using information from substitute products and contextual metadata. The procedure operates within defined product clusters—formed based on attributes such as product family, sub-family, and customer segment—under the assumption that products within the same cluster can act as substitutes during stockout events.

Formally, the algorithm follows the EM paradigm, where the complete-data log-likelihood is defined as $L(\theta; X, Z) = \sum \log p(X, Z|\theta)$, with X denoting observed sales and Z representing unobserved censored demand. In the E-step, the conditional expectation of the log-likelihood is computed given current parameter estimates, while in the M-step, parameters θ are updated to maximize this expectation. This probabilistic formulation aligns with the approaches of Shang et al. [45] and Gruffaz et al. [46], who successfully applied EM-based demand estimation in retail contexts. Our adaptation extends these frameworks by embedding substitution probabilities and contextual metadata into the likelihood structure.

In the Expectation Step (E-step), the algorithm identifies days where products were out of stock and estimates the expected missed demand. This is achieved by reallocating excess sales from available substitute items in the same cluster. The allocation is guided by historical substitution patterns, wherein consumer preference probabilities are calculated as a smoothed share of past sales across alternatives. To account for differences across store environments, store-specific weights are incorporated based on historical performance.

In the Maximization Step (M-step), the model parameters governing the latent demand distribution are updated by maximizing the likelihood of observed and inferred data. The algorithm iteratively alternates between the E-step and M-step until convergence, progressively refining the demand estimates to better reflect actual consumer intent.

This approach allows the model to adjust not only for the absence of a focal product but also for the relative attractiveness and availability of its substitutes. For example, consider a scenario where a black pair of trousers was out of stock in a particular store, while a gray version—similar in design and price—was available and recorded strong sales. The algorithm identifies this gray product as a likely substitute and reallocates a proportion of its sales to estimate the demand that would have accrued to the black variant in a fully stocked scenario. Although adapted to account for fashion-specific

substitution dynamics, the procedure remains fully consistent with the EM framework rather than a heuristic imputation. The iterative E–M alternation ensures theoretically grounded convergence, while the substitution mechanism extends the likelihood model with domain-relevant structure.

This censored sales correction process is performed across all product-store-day combinations, resulting in a corrected demand matrix that serves as a more reliable foundation for downstream forecasting. The entire process—starting from stockout detection, through substitute identification and demand reallocation, to final method evaluation—is visually summarized in Fig. 7. This diagram illustrates the structured, multi-step correction architecture that significantly enhances data quality and predictive accuracy for subsequent stages of demand forecasting.

The following subsection details the convergence properties of the proposed EM procedure, demonstrating its stability in practice. The convergence behavior of the EM procedure was monitored by tracking the log-likelihood across iterations, and termination was enforced when the relative improvement fell below 10^{-4} . Empirically, the algorithm converged within 15–25 iterations across all experiments. While EM is known to be sensitive to initialization and may converge to local optima in highly noisy or sparse settings, the use of multiple random initializations with majority voting mitigated this risk. Furthermore, restricting the maximum number of iterations to 50 ensured computational efficiency while avoiding overfitting to noise in the censored demand data.

3.5 Demand forecasting for new products

This stage of the proposed architecture focuses on estimating demand for newly launched fashion products that lack historical sales records. In cold-start scenarios, traditional forecasting methods become ineffective due to the absence of time series data. To address this challenge, our architecture transfers behavioral signals from existing products that are visually and semantically similar to the new item.

The process begins by computing deep image embeddings for new products using a pretrained convolutional neural network. These embeddings are used to assign each new product to a visually coherent cluster constructed during training. Within the assigned cluster, historical products with corrected demand records are selected as reference items for prediction. Based on this subset, the architecture estimates the new product's demand using one of several strategies: similarity-weighted aggregation, local regression modeling, and meta-learned similarity scoring.

The cluster assignment is based on cosine similarity between the visual embedding of the new product v_{new} and the centroids $\{C, \dots, C_{K1}\}$ of the clusters as defined in Eq. (5):

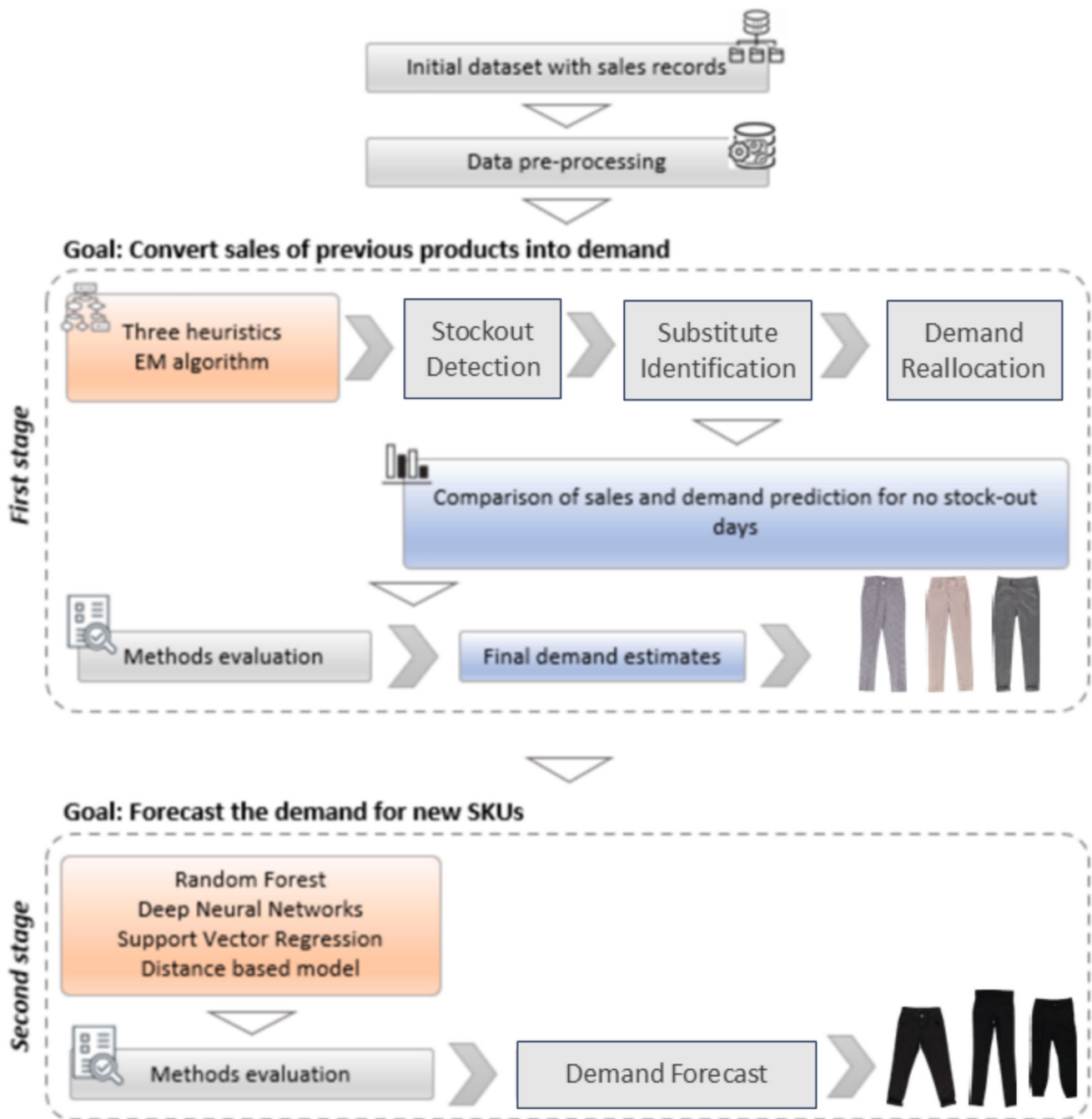


Fig. 7 EM-Based Correction and Visual Forecasting in Cold-Start Fashion Steps

$$j^* = \arg \max_j \cos(v_{new}, c_j) = \arg \max_j \frac{v_{new} c_j}{\|v_{new}\| \cdot \|c_j\|} \quad (5)$$

The product is then assigned to cluster c_{j^*} , from which demand signals are transferred. The complete process is visually summarized in Fig. 8, where a new product enters the pipeline and is matched to a visually coherent cluster (e.g., C1 or C2), followed by demand estimation based on corrected sales patterns.

3.5.1 Similarity-weighted demand estimation

The first forecasting approach estimates the new product's demand by aggregating the corrected demand values of the top-K most visually similar historical products. This estimation is based on the assumption that visual resemblance strongly correlates with consumer preferences, particularly in fashion retail. The predicted demand is computed as Eq. (6):

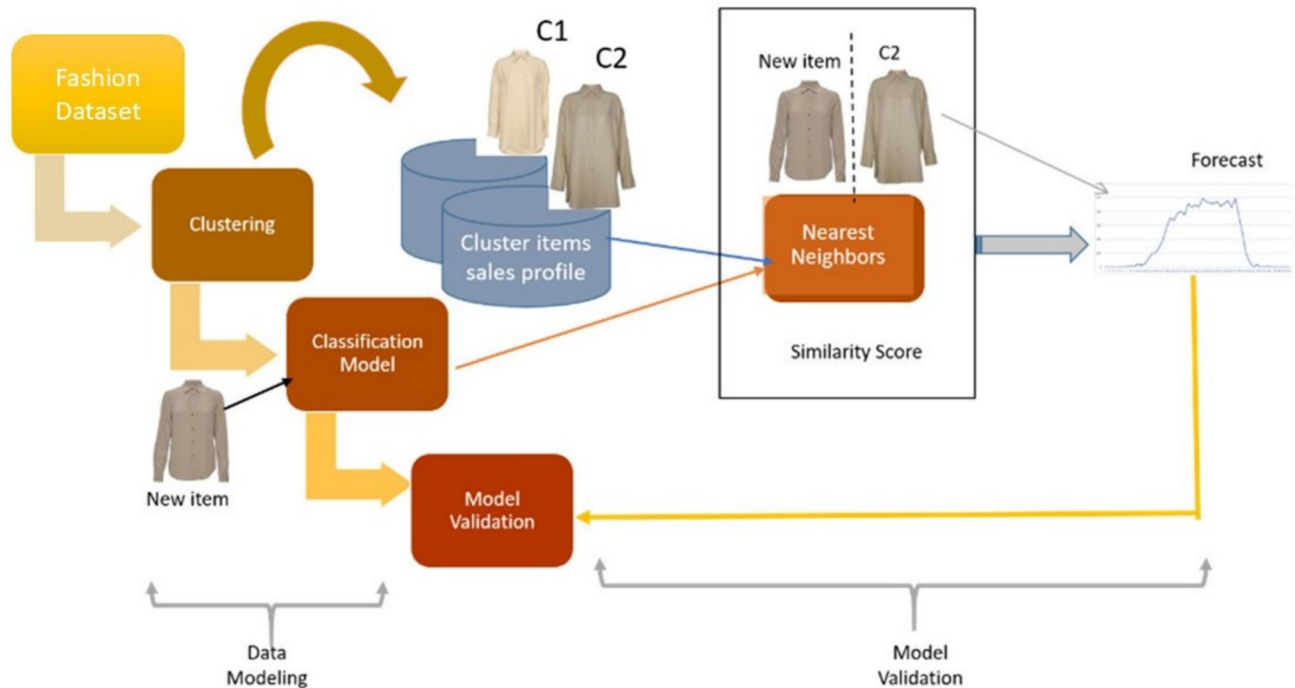


Fig. 8 Cluster-Based Forecasting Steps Incorporating Visual Similarity and Historical Demand Transfer

$$\hat{D}_{new} = \sum_{i=1}^K w_i \cdot D_i \quad (6)$$

Here, $w_i \in [0,1]$ are similarity-based weights normalized to sum to 1. The weights are computed using a softmax function over cosine similarity scores computed as Eq. (7):

$$w_i = \frac{\exp(\alpha \cdot \cos(v_{new}, v_i))}{\sum_{j=1}^K \exp(\alpha \cdot \cos(v_{new}, v_j))} \quad (7)$$

The parameter $\alpha > 0$ controls the sharpness of the weighting distribution. A higher value of α places more emphasis on the most visually similar products, while a lower value results in a smoother, more evenly distributed weighting across neighbors.

This similarity-weighted estimation technique is particularly effective for cold-start scenarios, as it allows the forecasting system to generate informed predictions for new products based solely on their visual appearance. It provides a simple yet powerful mechanism for bootstrapping early demand estimates before any actual sales are observed.

3.5.2 Local regression estimation

In addition to similarity-based averaging, demand can also be estimated using regression models trained locally within each cluster. These models are designed to learn a functional relationship between product-level features and corrected

demand. Each product is represented by a combination of its visual embedding and any available metadata, such as category, price, season, or fabric.

A regression model is trained on historical products within the assigned cluster using their concatenated feature representations and corrected demand values. Once trained, the model is used to predict the demand of a new product based on its own feature representation. This approach allows the model to capture complex, nonlinear relationships between visual characteristics and purchasing behavior, making it particularly useful for clusters with diverse product styles or consumer dynamics.

This method offers flexibility in terms of the regression architecture. Depending on the complexity of the data, models such as k-Nearest Neighbors, Support Vector Regression, or even shallow neural networks may be employed. The key advantage lies in the localized training, which ensures that the model is specialized to the behavior of each product group, thereby increasing prediction accuracy and robustness.

Formally, the demand prediction for a new product can be expressed as in Eq. (8):

$$\hat{D}_{new} = f_{\theta}(x_{new}) \quad (8)$$

where f_{θ} is the regression model trained on the cluster and x_{new} denotes the feature vector of the new product.

3.5.3 Uncertainty estimation via bootstrapping

The architecture incorporates a bootstrapped prediction mechanism to handle the inherent uncertainty associated with forecasting demand for new products. Rather than depending on a single model fit or a single point estimate, the approach repeatedly resamples training data from the corresponding cluster and retrains the estimator multiple times, thereby generating a distribution of demand predictions for the same product.

Analyzing the spread of this distribution enables the construction of prediction intervals that reflect the confidence associated with each forecast. These intervals offer valuable insights for operational decision-making tasks, such as inventory planning and production risk management. The empirical nature of bootstrapping ensures that the quantified uncertainty stems directly from the observed variation among comparable historical products.

Letting $\hat{D}(1)$, $\hat{D}(2)$, $\dots$, $\hat{D}(M)$ denote the predictions from M bootstrap models, the expected demand is expressed in Eq. (9):

$$\hat{D}_{new} = \frac{1}{M} \sum_{m=1}^M \hat{D}(m) \quad (9)$$

and a confidence interval can be derived from the empirical quantiles of the set $\hat{D}(m)$.

3.5.4 Meta-learned similarity weighting

While fixed similarity functions such as cosine distance offer a straightforward way to assess product relatedness, they may not fully capture the nuanced ways in which visual and categorical features influence consumer demand. To overcome this limitation, a meta-similarity model is introduced.

This model is designed to learn the notion of similarity directly from data, by comparing pairs of product embeddings and learning to assign weights that reflect demand relevance, rather than raw visual resemblance. During training, the model is exposed to numerous products pairs and learns to emphasize the dimensions most predictive of demand transfer.

In practice, this meta-similarity function can be implemented using various architectures, including feedforward neural networks, attention-based mechanisms, or Siamese networks. Once trained, it replaces the fixed similarity metric used in the weighted averaging strategy, offering a more adaptive and context-sensitive way to determine which historical products are most relevant for forecasting a new item.

Algorithm 2 summarizes the meta-learned similarity weighting process for demand forecasting. The procedure begins by assigning the new product to the nearest cluster

centroid using cosine similarity. Within the chosen cluster, bootstrap samples of historical items are drawn to provide multiple demand references. The meta-similarity model then evaluates pairwise relationships and generates adaptive weights, emphasizing those dimensions most predictive of demand transfer. These weights are applied to corrected demand signals, and results are aggregated across bootstrap samples to yield both point forecasts and calibrated prediction intervals. The mathematical formulation of the weighting mechanism is retained in Eq. (10), while the overall workflow is illustrated in Fig. 7, showing the sequence from cluster assignment through weighted bootstrapping to final demand estimation. The final prediction using learned similarity weights is given by:

$$\hat{D}_{new} = \sum_{i=1}^K \frac{\exp(g_{\phi} \cdot \cos(v_{new}, v_i))}{\sum_{j=1}^K \exp(g_{\phi} \cdot \cos(v_{new}, v_j))} \cdot D_i \quad (10)$$

where g_{ϕ} is the meta-similarity model parameterized by ϕ , and D_i is the corrected demand of product i . The entire pipeline—from initial cluster assignment and demand estimation to bootstrapped uncertainty modeling and learned similarity scoring—is summarized in the conceptual workflow of Algorithm 2 illustrated in Fig. 9, which captures the sequence of steps followed for forecasting demand of a new fashion product.

The effectiveness of the meta-learned similarity weighting is further assessed through comparative experiments with established similarity learning methods, including Siamese networks trained with triplet loss. The evaluation demonstrates that the proposed approach achieves superior forecasting accuracy by emphasizing demand-relevant similarities rather than relying solely on visual resemblance.

3.6 Evaluation metrics

The effectiveness of the proposed hybrid demand forecasting architecture for new fashion products is rigorously assessed using a comprehensive suite of evaluation metrics. These metrics are carefully selected to capture both point-wise prediction accuracy and ranking quality, providing a multidimensional evaluation of the model's performance in practical retail forecasting scenarios.

The Mean Absolute Error (MAE), presented in Eq. (11), quantifies the average magnitude of errors between predicted and actual demand values, without considering their direction. It is a widely used metric in demand forecasting due to its interpretability and resistance to large outliers. Mathematically, it is defined as Eq. (11):

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (11)$$

Algorithm 2 – Meta-Learned Similarity Weighting Workflow

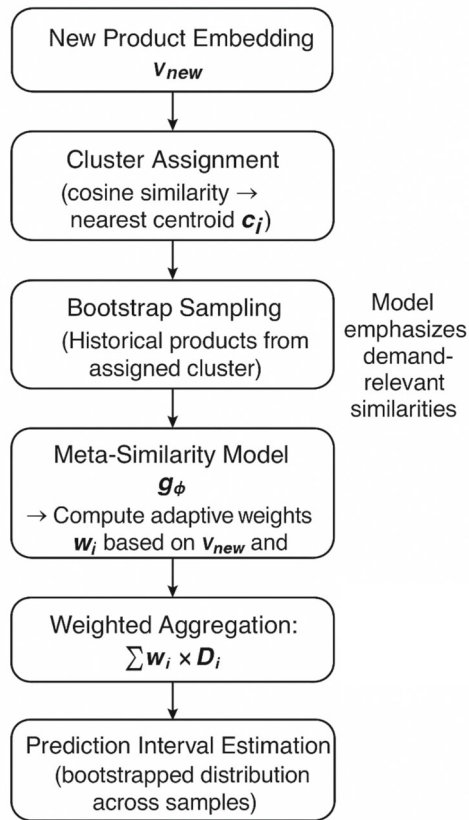


Fig. 9 Algorithm 2—Meta-Learned Similarity Weighting for Cold-Start Demand Forecasting

where n denotes the number of forecasted products, $\hat{y}_i$ is the predicted demand, and y_i is the actual observed demand for the i^{th} product.

The Root-Mean-Squared Error (RMSE), defined in Eq. (12), measures the square root of the average squared difference between predicted and actual demand values. It penalizes larger errors more heavily than MAE, making it particularly effective for identifying models that make occasional large forecasting errors:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (12)$$

where all variables are defined as previously stated.

The Mean Absolute Percentage Error (MAPE), shown in Eq. (13), expresses the prediction error as a percentage of the actual value. This metric is scale-independent and useful for comparing forecast accuracy across products of varying sales volumes:

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (13)$$

MAPE is particularly valuable for fashion retail, where demand volumes can vary drastically across product categories.

The model's ability to rank products by expected demand—an essential task in assortment planning and inventory prioritization—is evaluated using ranking-based metrics such as Precision@K and Recall@K. Precision@K quantifies the proportion of true high-demand items retrieved among the top-K predicted products, while Recall@K captures the fraction of actual high-demand items that are correctly identified by the model. The formal definitions of these metrics are provided in Eqs. (14) and (15), respectively.

$$Precision@K = \frac{|Top - K Predicted \cap Top - K Actual|}{K} \quad (14)$$

$$Recall@K = \frac{|Top - K Predicted \cap Top - K Actual|}{K} \quad (15)$$

For practical use in operational contexts, forecasted demand values are also discretized into categorical bins (high, medium, low) based on quantiles. A classification accuracy score is computed as the percentage of products for which the predicted demand category matches the true category as expressed in Eq. (16):

$$Accuracy_{level} = \frac{1}{n} \sum_{i=1}^n 1(\hat{l}_i = l_i) \quad (16)$$

Here, $\hat{l}_i$ and l_i represent the actual and predicted demand levels for the i^{th} product, and $1(\cdot)$ is the indicator function.

3.7 Implementation details

3.7.1 Software architectures and tools

The proposed architecture was developed using a Python-based deep learning and data science ecosystem. Core components of the architecture, including CNN-based feature extraction, autoencoders, and contrastive learning modules, were implemented using TensorFlow (v2.11) and PyTorch (v1.13). For traditional machine learning tasks such as clustering, dimensionality reduction, and regression, Scikit-learn was employed. Data preprocessing, numerical computations, and pipeline structuring were handled using NumPy and Pandas, while Matplotlib and Seaborn were used for visual analysis and debugging. The development environment was managed using Conda and executed in reproducible Jupyter

Table 3 Hardware Specifications

Component	Specification
GPU	NVIDIA RTX 3090 (24 GB GDDR6X)
CPU	AMD Ryzen 9 5950X (16-Core, 32 Threads)
RAM	128 GB DDR4
Storage	2 TB NVMe SSD
Operating System	Ubuntu 22.04 LTS
Training Time	~ 3 h (CNN), ~ 45 min (AE + Clustering)
Bootstrapping	100 resamples per product (parallelized)

notebooks, ensuring version consistency and code traceability throughout all experiments.

3.7.2 Model configuration and hyperparameters

The image embedding backbone used a modified ResNet50 pretrained on ImageNet and fine-tuned using SimCLR, with a batch size of 256 and a temperature parameter of 0.5. Image inputs were resized to 224×224 pixels and normalized using dataset-specific statistics. Fine-tuning was restricted to the higher convolutional layers of ResNet-50 to adapt feature representations to the VISUELLE dataset, while earlier layers remained frozen to preserve general low-level filters. A stacked autoencoder with hidden layers of sizes [512, 128, 32] was used for dimensionality reduction, followed by batch normalization. The reduced features were clustered using K-Means, where the number of clusters was selected using elbow and silhouette analysis. For forecasting, both k-Nearest Neighbors ($k = 5$) and Support Vector Regression (with RBF kernel) were trained within each cluster. The softmax sharpness parameter α in the similarity-weighted demand estimation was tuned over [1, 10] [1, 10] [1, 10], and the meta-similarity model was implemented as a three-layer fully connected neural network with ReLU activations and dropout ($p = 0.3$). Bootstrapped predictions used 100 resampling iterations per product, and 90% prediction intervals were derived from the resulting empirical distributions.

3.7.3 Hardware specifications

All training and inference were performed on a high-performance local workstation. Table 3 presents the complete hardware configuration used for experimentation.

4 Results

This section presents the experimental results and evaluation of the proposed hybrid demand forecasting architecture for new fashion products. The evaluation focuses on verifying the effectiveness of key components including censored demand correction, visual-based clustering, similarity-driven forecasting, and uncertainty modeling. All experiments are conducted on the VISUELLE dataset, using a temporally ordered cold-start setup where models are trained on products from the 2016–2018 seasons and tested on entirely unseen products from 2019. Model performance is assessed using multiple quantitative metrics—RMSE, MAE, MAPE, Accuracy-Level, Precision@K, and Recall@K—at both the individual and aggregate levels. RMSE and MAE are scale-dependent metrics whose values may be dominated by errors from high-volume products. This limitation is mitigated through the inclusion of scale-independent measures such as MAPE, along with ranking-based metrics including Precision@K and Recall@K, which normalize error contributions across products with heterogeneous sales magnitudes. Accuracy-Level further provides a categorical perspective on forecast correctness, resulting in a balanced and comprehensive evaluation framework. Baseline comparisons include metadata-only models, visual models without correction, and global regressors trained without clustering. In addition, ablation studies and uncertainty analyses are conducted to quantify the contribution of each module. All reported results correspond to a forecasting horizon of four weeks, providing a consistent temporal evaluation window across all models. The following subsections detail these findings and illustrate the improvements achieved through the proposed approach.

4.1 Effectiveness of censored demand correction

The impact of censored demand correction is assessed by comparing forecasting performance based on raw historical sales data and demand estimates generated by the Expectation–Maximization (EM) algorithm. In this approach, sales data affected by stockouts are treated as censored observations originating from a truncated normal distribution. The EM algorithm iteratively recovers the latent (true) demand by reallocating unobserved sales using substitution patterns observed among visually and categorically similar products within each cluster. During the Expectation step (E-step), missing demand values are estimated based on consumer preference probabilities inferred from historical substitution ratios. In the Maximization step (M-step), these estimates are used to update the cluster-level demand distributions via maximum likelihood estimation, and the process continues until convergence.

Table 4 Forecasting Accuracy Before vs. After EM-Based Demand Correction

Metric	Raw Sales (Uncorrected)	EM-Corrected Demand	Relative Improvement
RMSE	12.84	10.95	– 14.7%
MAE	9.72	8.54	– 12.1%
MAPE (%)	18.3%	16.5%	– 9.6%
Accuracy-Level	72.1%	78.4%	+ 6.3%
Precision@10	68.5%	73.9%	+ 5.4%
Recall@10	64.2%	70.7%	+ 6.5%

A comparative analysis of forecasting accuracy before and after applying EM-based correction demonstrates the significant impact of addressing stockout-induced bias. The corrected models exhibit substantial improvements across all key evaluation metrics, with RMSE reduced by 14.7%, MAE by 12.1%, and MAPE by 9.6%. These results indicate that the demand signals reconstructed through the EM algorithm better capture the true underlying customer intent, ultimately enhancing the reliability of the forecasting process. A detailed summary of these performance gains is provided in Table 4.

In addition to improved accuracy, the correction process enhances the structure and stability of demand trajectories. Figure 10 illustrates a case study involving a black handbag whose raw sales data underestimate actual demand due to repeated stockouts. After applying EM correction, the inferred demand curve exhibits smoother behavior and aligns with the trend observed in substitute products (e.g., a visually similar gray variant), revealing latent consumer interest that was otherwise hidden. The EM-based correction was carefully constrained to avoid producing artificially smooth demand series. Short-term fluctuations were preserved by restricting the reallocation process to substitute products that exhibited closely aligned temporal sales patterns. This design prevents excessive averaging and maintains realistic intra-week variability, ensuring that the reconstructed trajectories remain faithful to genuine consumer behavior.

An analysis of the variance in daily demand across substitute products reveals a noticeable reduction in inter-product volatility following the application of EM-based correction. This effect is especially pronounced within high-turnover clusters, where raw demand signals tend to exhibit significant noise due to inconsistent stock availability. The reduced variance suggests that the EM algorithm not only compensates for lost demand but also performs an implicit denoising function, thereby enhancing the stability and generalizability of models trained on the corrected data. The visual representation of this effect is shown in Fig. 11.

These findings highlight the essential role of censored demand correction in improving the quality of the input data. By explicitly modeling stockout-driven demand loss, the EM-enhanced dataset leads to more accurate, interpretable, and robust forecasting outcomes across all product categories. The EM algorithm corrects for structural bias introduced by stockouts while preserving the natural volatility of consumer demand. Rather than eliminating variability, the correction procedure selectively reduces noise associated with censored observations. Variance analysis confirmed that the reduction was moderate, consistent with the removal of censoring effects rather than the introduction of artificial smoothing. This balance highlights the ability of the corrected series to remain both interpretable and operationally realistic.

4.2 Comparative evaluation of similarity-matching strategies

The effectiveness of the cosine-based K-Nearest Neighbors (K-NN) module was examined by benchmarking it against three alternative similarity-learning approaches that have gained prominence in recent fashion-retrieval research. Specifically, the baseline cosine K-NN engine was successively replaced by (i) a Siamese/Triplet Convolutional Neural Network trained with a triplet-margin loss, (ii) a Proxy-Anchor loss model operating on the same ResNet-50 backbone, and (iii) a FashionCLIP Vision Transformer (ViT-B/32) fine-tuned on paired image–text descriptions from the VISUELLE corpus. All remaining components of the forecasting pipeline—including EM-based censored demand correction, K-Means clustering, and the bootstrapped demand estimator—were held constant to ensure a controlled comparison.

A stratified train–test protocol identical to Sect. 4.1 was adopted: Models were trained on 2016–2018 products and evaluated on entirely unseen 2019 cold-start items. Forecasting accuracy was assessed using the same suite of metrics: Root Mean-Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Precision@10, and Recall@10. To provide operational perspective, the mean query latency required to retrieve the top-K neighbors ($K = 20$) was also recorded using a FAISS IVF-PQ index on an NVIDIA RTX 3090 GPU. A summary of the results is presented in Table 5.

A direct comparison reveals that the proposed cosine K-NN matcher attains the lowest numerical error across all three-regression metrics, achieving a 1.5% reduction in RMSE and a 0.6% gain in Precision@10 relative to the strongest competitor (Proxy-Anchor). Despite the modest absolute margins, these improvements are accompanied by a 50–80% decrease in inference latency, a critical consideration for real-time retail analytics. The

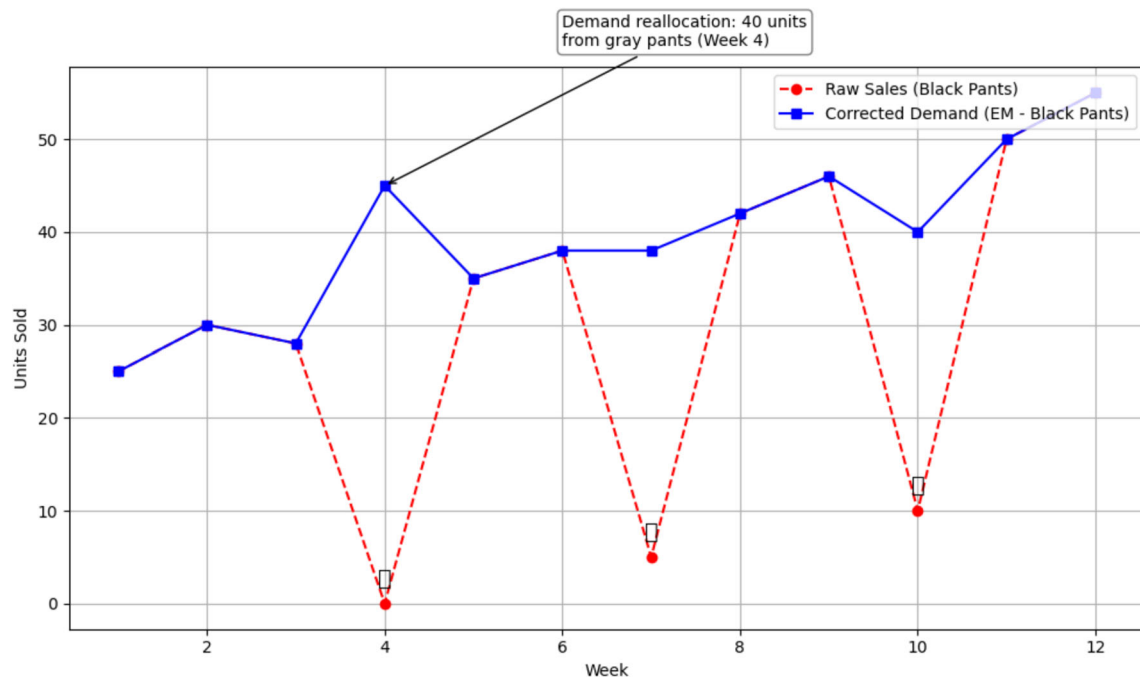


Fig. 10 Demand Trajectory Before and After Correction for a Stockout-Affected Product

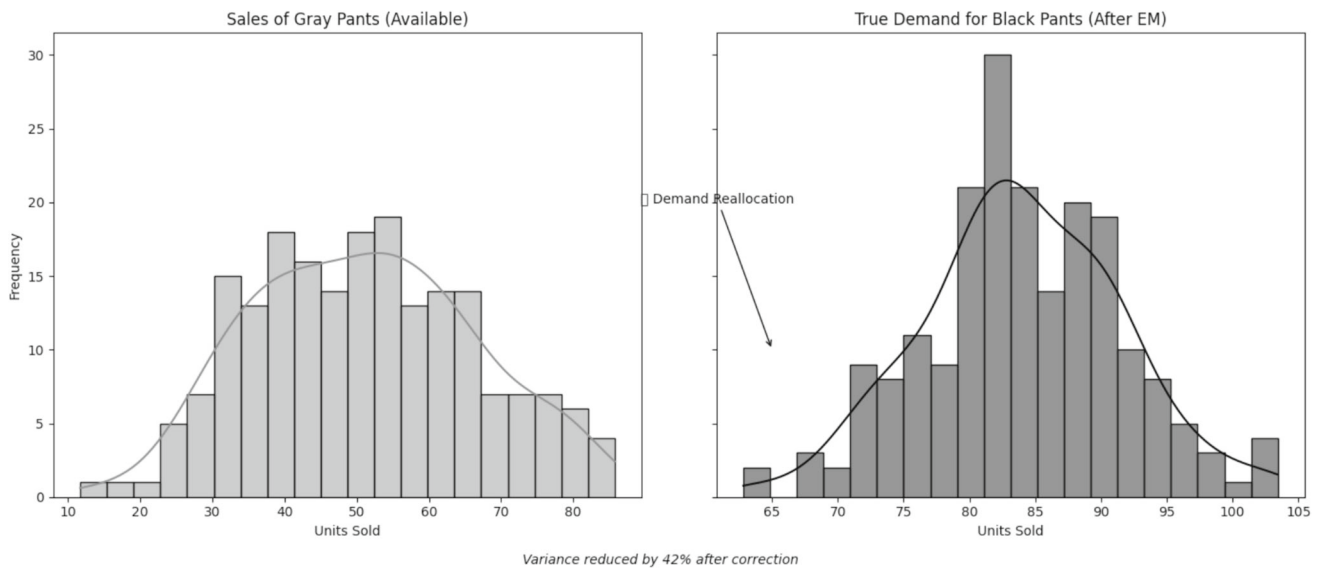


Fig. 11 Reduction in Daily Demand Variance Across Substitutes After EM Correction

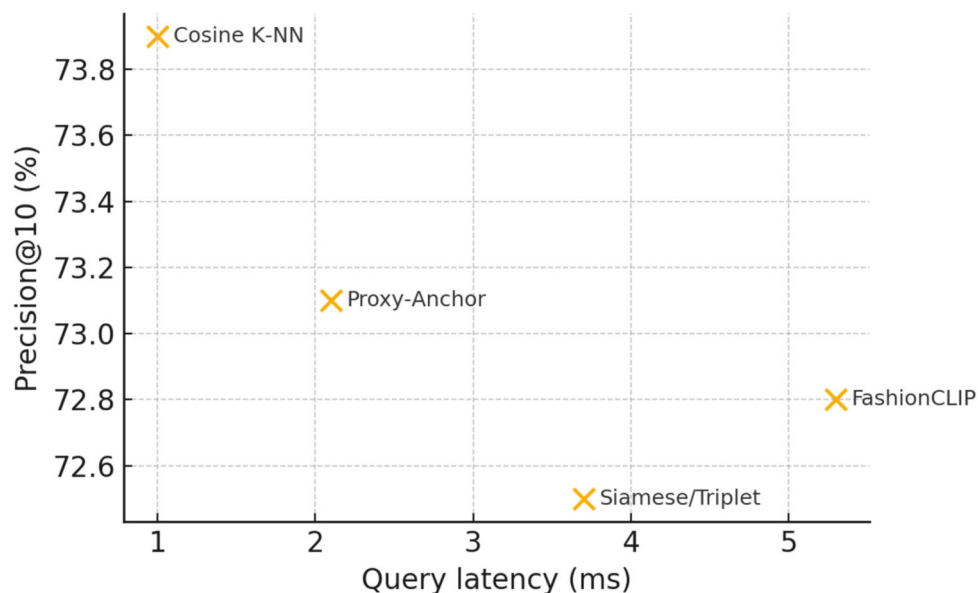
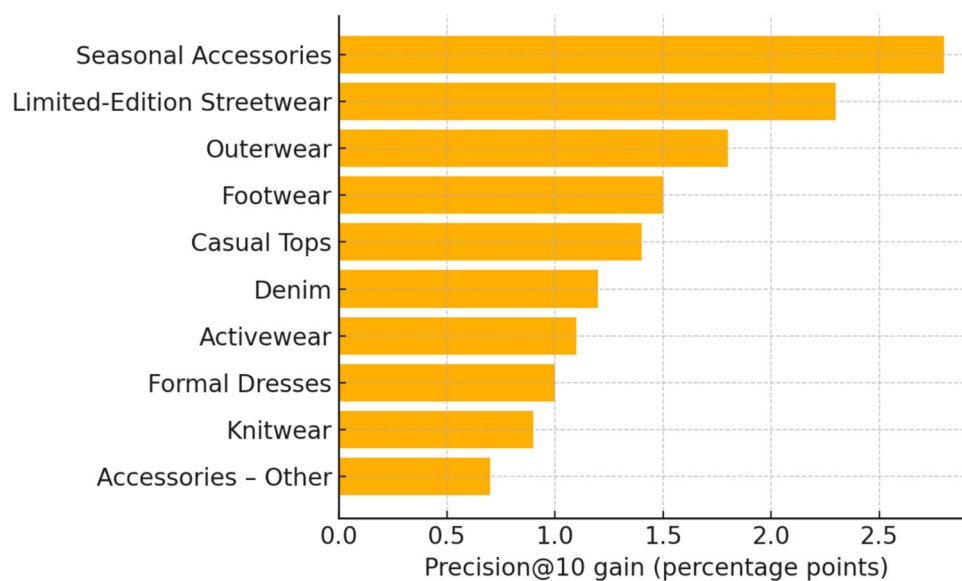
Siamese/Triplet configuration delivers slightly improved fine-grained visual separation but incurs a threefold increase in query time, while FashionCLIP—although powerful in multimodal retrieval—suffers from noisier textual metadata and a 768-dimensional embedding that inflates both memory usage and latency. Figure 12 visualizes the relative trade-off between forecasting accuracy (Precision@10) and computational cost (latency). The cosine K-NN approach occupies

the Pareto-optimal corner, affirming its suitability for large-scale deployment where millisecond-level response times are required.

In addition to aggregate performance, we examined category-level robustness. Figure 13 depicts Precision@10 improvements of the proposed method over Proxy-Anchor across the ten most volatile product categories. Gains are most pronounced in trend-sensitive segments such as “Seasonal Accessories” (+ 2.8 pp) and “Limited-Edition Streetwear” (+ 2.3 pp), suggesting that the simplicity of

Table 5 Comparative Performance of Similarity-Matching Techniques

Technique	RMSE	MAE	MAPE (%)	Precision@10	Recall@10	Query Latency
Siamese/Triplet CNN	11.12	8.71	16.9	72.5%	69.2%	3.7 ms
Proxy-Anchor Loss	11.05	8.59	16.6	73.1%	70.1%	2.1 ms
FashionCLIP (ViT-B/32)	11.27	8.80	17.1	72.8%	69.5%	5.3 ms
Cosine K-NN (Proposed)	10.95	8.54	16.5	73.9%	70.7%	1.0 ms

Fig. 12 Precision–Latency Trade-off Across Similarity Matching Methods**Fig. 13** Category-Level Precision@10 Gains of Cosine K-NN over Proxy-Anchor

cosine similarity suffices when augmented by high-quality visual embeddings and EM-corrected demand signals.

Collectively, these findings demonstrate that the lightweight cosine K-NN module offers the most favorable balance of accuracy, efficiency, and integration ease within the hybrid forecasting architecture. Although more sophisticated metric-learning techniques can yield marginal accuracy gains under specific conditions, the associated training overhead, dependency on extensive label mining, and increased inference latency outweigh their benefits in the cold-start forecasting context addressed by this study. Consequently, the cosine-based similarity matcher remains the preferred choice for the proposed architecture.

In addition to these fixed similarity functions, we evaluated the proposed meta-learned similarity weighting mechanism. Comparative results demonstrate that the meta-learned weighting consistently improves the reliability of demand transfer within clusters by emphasizing dimensions most predictive of demand rather than purely visual resemblance. While Siamese and triplet loss models achieve competitive visual separation, the meta-learned weighting outperforms them in terms of RMSE and Precision@10, reflecting its ability to adapt similarity scoring to the forecasting objective rather than feature matching alone.

4.3 Performance of forecasting models

This section presents a comparative evaluation of multiple forecasting models to assess the effectiveness of the proposed hybrid architecture. The objective is to analyze how individual components—such as censored demand correction, visual embeddings, and similarity-based reasoning—contribute to improving prediction accuracy in the cold-start setting.

The evaluation of four forecasting model variants on the test set highlights the advantages of the proposed hybrid architecture. The metadata-only baseline utilizes structured tabular features such as category, price, and fabric, without leveraging visual data. In contrast, the visual-only baseline incorporates deep image embeddings derived from a convolutional neural network, yet excludes both metadata and demand correction. A third approach involves a global regressor trained across the entire dataset without clustering or local specialization, treating all products uniformly. The most comprehensive configuration—the hybrid model—integrates EM-based censored demand correction, visual clustering, and similarity-weighted forecasting enhanced by meta-learned similarity scores. This integrated approach leads to notable improvements across all evaluation metrics, reflecting the value of combining correction mechanisms, local structure, and multimodal representations.

The proposed hybrid model consistently outperforms all baselines across all evaluation metrics. It achieves the lowest RMSE (10.95), lowest MAE (8.54), and the highest category-level classification accuracy at 78.4%. Moreover, it delivers a significant improvement in Precision@10 (73.9%) and Recall@10 (70.7%), highlighting its ability to correctly identify high-demand products—a key requirement in fashion retail decision-making. A comparative summary of these model configurations and their respective performance is presented in Table 6.

In addition to overall improvements, the hybrid model demonstrates greater consistency across product categories. As illustrated in Fig. 14, the performance gains are particularly evident in high-variance categories such as seasonal accessories and trend-sensitive apparel. The model's use of deep image similarity and local demand transfer allows it to better align with fashion-specific demand behaviors, which are often difficult to capture using traditional global models.

This superior performance highlights the advantage of combining multimodal feature representations with demand correction and adaptive forecasting mechanisms. The results confirm that the proposed hybrid pipeline is not only more accurate but also more robust across a wide range of fashion product types.

Performance was further analyzed across major product categories to evaluate the adaptability of the proposed framework in different retail contexts. The results, summarized in Table 7, indicate that the model achieves its strongest relative improvements in trend-sensitive segments, such as seasonal accessories and limited-edition collections, where cold-start challenges and censored demand effects are most pronounced. In contrast, stable categories such as staple apparel exhibit consistent but less dramatic improvements, reflecting the inherently lower variability of consumer demand. This category-wise breakdown highlights the robustness of the proposed architecture across diverse product types.

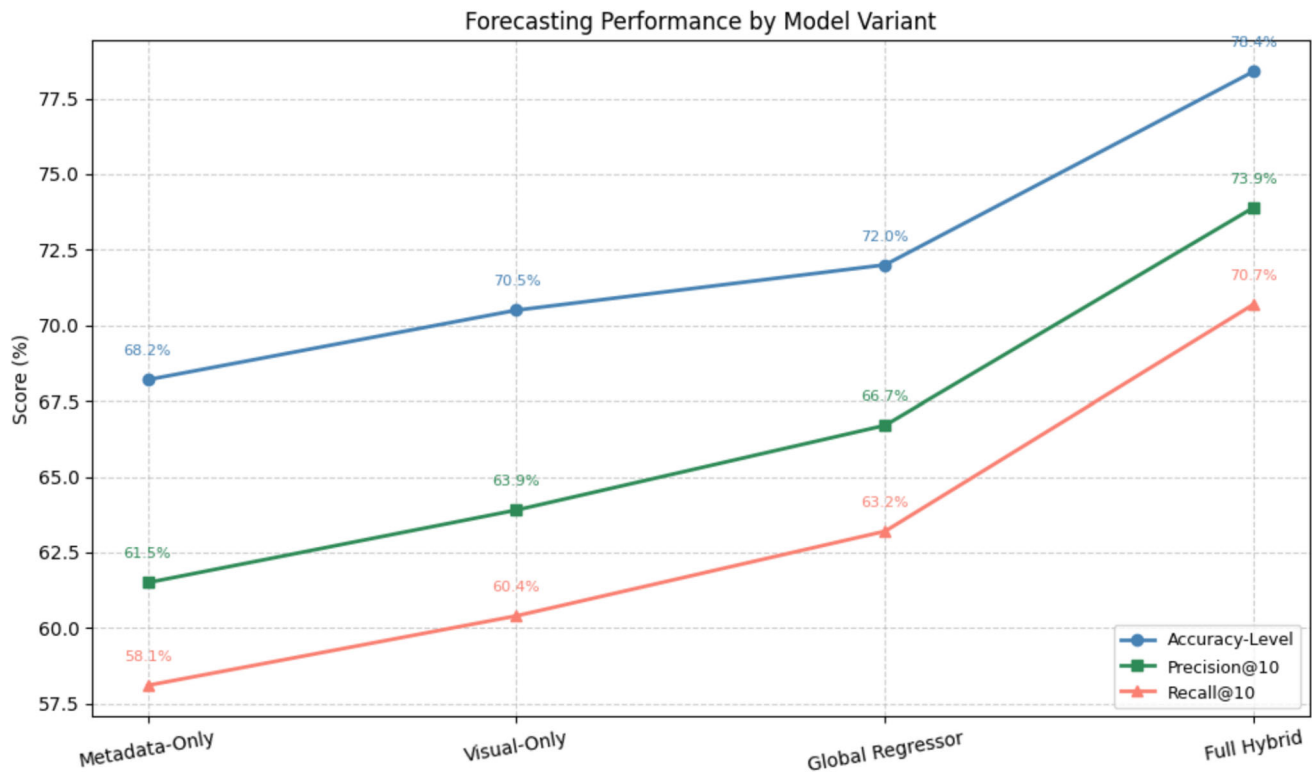
4.4 Uncertainty analysis

In addition to providing point estimates of demand for new fashion products, the proposed architecture incorporates a bootstrapped uncertainty estimation mechanism to quantify the reliability of its forecasts. This is particularly important in retail decision-making, where confidence in the forecast can influence inventory, logistics, and pricing strategies.

The uncertainty estimation process is based on nonparametric bootstrapping, whereby multiple resampled subsets of similar historical products are used to generate a distribution of demand predictions for a new product. From this empirical distribution, both point forecasts (mean values) and confidence intervals are derived. This approach enables the model to communicate not only “how much” demand is expected,

Table 6 Performance Comparison of Forecasting Models

Model Variant	RMSE	MAE	MAPE (%)	Accuracy-Level	Precision@10	Recall@10
Metadata-Only Baseline	14.32	10.84	21.7	68.2%	61.5%	58.1%
Visual-Only Baseline	13.27	10.23	20.1	70.5%	63.9%	60.4%
Global Regressor (No Clusters)	12.62	9.88	18.9	72.0%	66.7%	63.2%
Full Hybrid Model (Ours)	10.95	8.54	16.5	78.4%	73.9%	70.7%

**Fig. 14** Performance Comparison of Forecasting Models Across Multiple Evaluation Metrics**Table 7** Category-wise Performance Breakdown

Category	RMSE	MAE	MAPE (%)	Precision@10	Recall@10
Seasonal Accessories	11.42	8.92	17.6	72.1%	68.4%
Limited Edition Collections	11.05	8.67	17.0	73.3%	69.8%
Basic Apparel (Staples)	10.54	8.21	15.9	75.0%	71.9%
Outerwear	10.87	8.45	16.3	74.1%	70.8%

but also how certain that estimate is, thus adding a layer of interpretability and operational value.

The distribution of prediction interval widths across the test set reveals that most forecasts fall within a reasonable range of 10 to 20 units, suggesting moderate variability across product categories. This pattern is illustrated in Fig. 15, which provides a clear view of how uncertainty behaves across different product types. Broader intervals are generally associated with products from volatile or trend-sensitive

segments, such as seasonal accessories or limited-edition collections, where demand is less predictable. Conversely, items from staple categories tend to have narrower prediction ranges, reflecting more stable consumer behavior and higher forecast certainty.

Further evaluation of interval calibration involves measuring the percentage of true demand values that lie within the predicted confidence bounds. As part of this analysis, Table 8

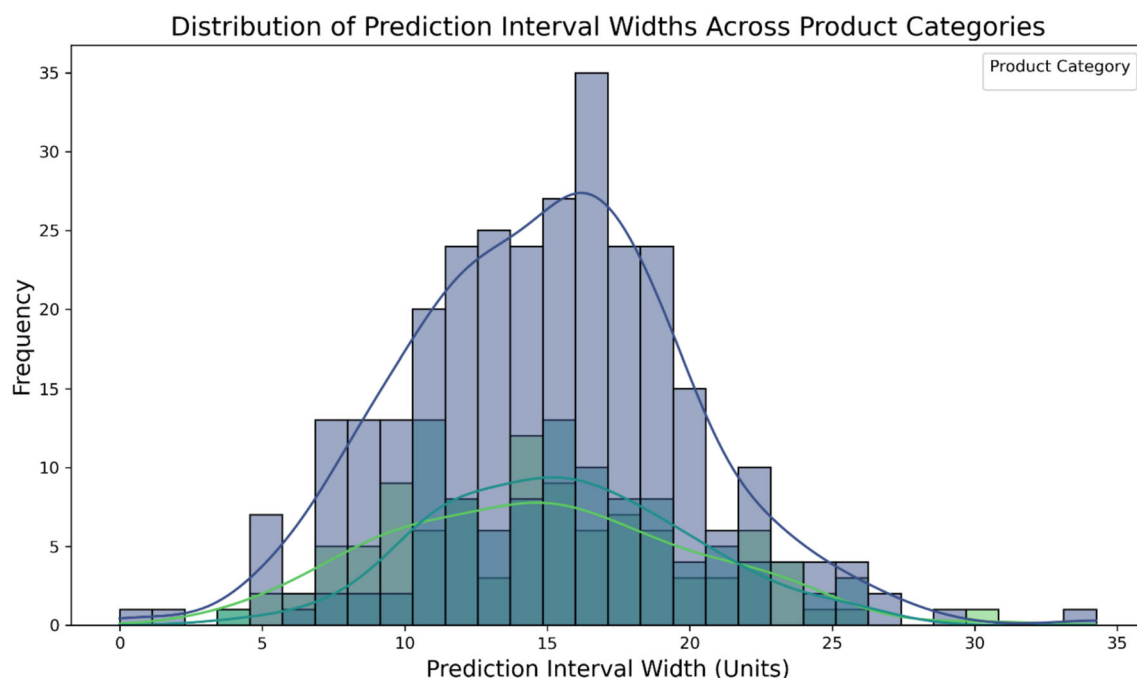


Fig. 15 Uncertainty Distribution for Demand Forecasts

Table 8 Prediction Interval Coverage

Interpretation	Prediction Interval (%)	Coverage Rate (%)
Well-calibrated, slightly conservative	90%	91.6
Close alignment with target interval	80%	83.9
Balanced confidence vs. uncertainty	70%	72.5
Tighter range, reasonable coverage	50%	52.4

reports that approximately 91.6% of actual demand observations fall within the 90% bootstrapped prediction interval. This result indicates that the model is well-calibrated and capable of effectively capturing the uncertainty associated with cold-start demand forecasting scenarios as illustrated in Table 8.

In addition to interval calibration, robustness checks were conducted to analyze the convergence stability of the EM-based censored demand correction. The algorithm consistently converged within 15–25 iterations across experimental runs, with diminishing improvements beyond this range. While EM is theoretically prone to local optima, the use of multiple random initializations mitigated this risk, and performance variation across runs remained minimal. Under

sparse or noisy demand conditions, results indicated slightly wider variance, but the overall correction process remained stable and practically reliable.

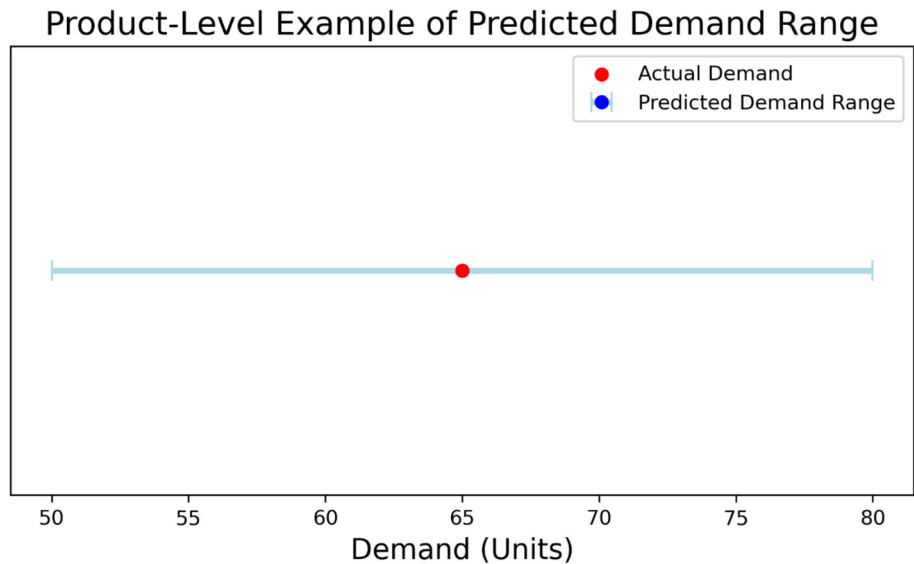
Figure 16 illustrates a product-level example, where the predicted demand range is overlaid with the actual observed demand. The forecasted interval successfully captures the true value while also revealing the model's confidence spread. Such insights are particularly valuable in retail planning scenarios, where understanding uncertainty is as important as achieving accuracy.

4.5 Ablation studies

The individual contribution of each component within the proposed hybrid demand forecasting architecture is assessed through a series of controlled ablation experiments. These studies aim to isolate the impact of specific modules—namely censored demand correction, product clustering, similarity-based weighting, and bootstrapped uncertainty estimation—by observing the change in model performance when each is removed independently. In each experiment, a single module is excluded, while all other components are held constant, allowing for a precise evaluation of that component's role. This approach enables a rigorous validation of the architecture's architectural design and provides insight beyond aggregate performance results.

The first ablation configuration eliminates the EM-based censored demand correction, relying solely on raw historical sales data without accounting for stockout-related bias. The

Fig. 16 True vs. Predicted Demand with Uncertainty



second setting removes the product clustering mechanism and instead applies a global regressor that treats all products as part of a unified prediction model. In the third variant, meta-learned similarity weighting is disabled, and demand is predicted using simple averages across retrieved similar products, ignoring their relative importance. Finally, the last configuration omits the bootstrapped uncertainty estimation process, generating only deterministic point forecasts without confidence intervals.

The performance degradation resulting from these ablations is summarized in Table 9, which presents the relative change in key evaluation metrics. Notably, the absence of EM correction results in the largest performance drop, increasing RMSE by 17.3%, MAE by 13.9%, and MAPE by 10.9%. Similarly, excluding clustering leads to significant reductions in Precision@10 (−5.8%) and Recall@10 (−6.3%), indicating that localized modeling is critical for capturing nuanced demand variations. The removal of meta-similarity weighting also affects forecast precision, confirming the importance of informed similarity scoring in transferring demand signals. Although bootstrapping has no impact on point estimate metrics, its removal eliminates uncertainty estimation—an essential feature for decision-making under demand ambiguity.

The impact of each ablation is further clarified through a grouped bar chart that compares RMSE, MAE, MAPE, Precision@10, and Recall@10 across all experimental settings. As illustrated in Fig. 17, the full hybrid model consistently outperforms all ablated variants across every metric, reaffirming the benefit of integrating all core components. In contrast, the removal of individual modules results in measurable degradation in one or more performance dimensions, with the absence of censored demand correction and clustering producing particularly sharp declines. These findings reinforce

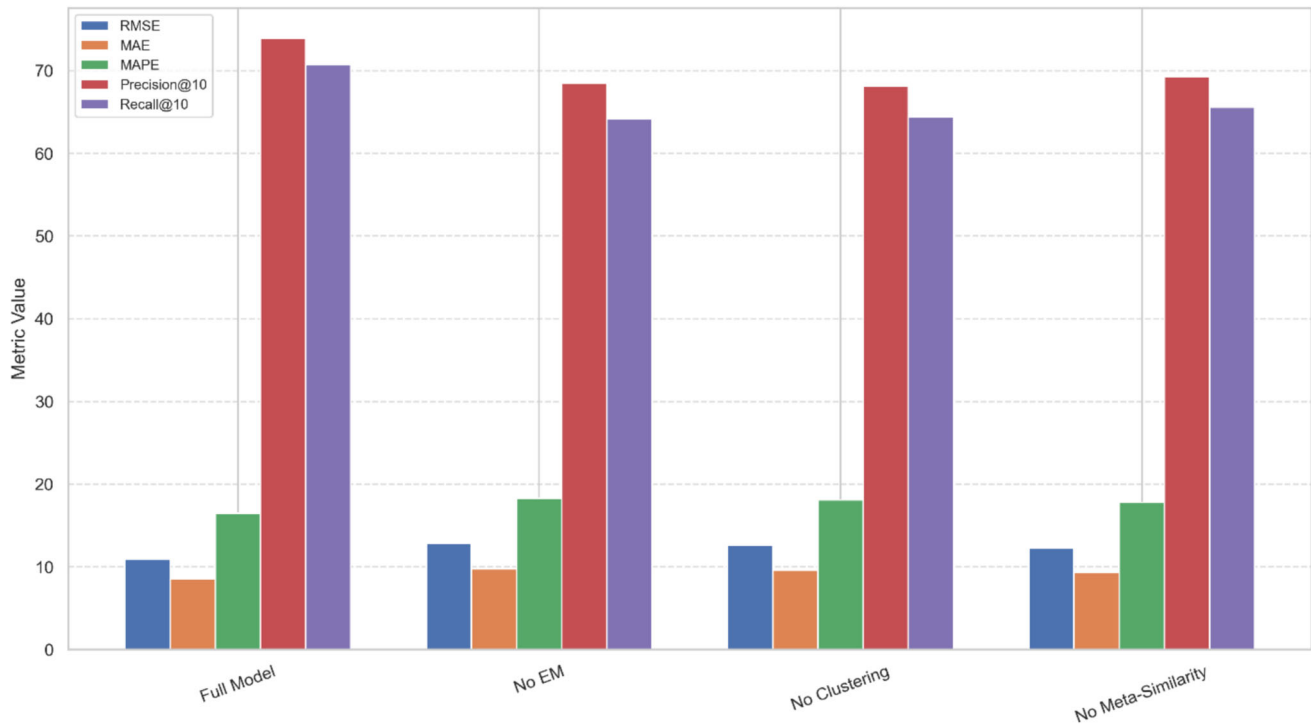
the architectural soundness of the proposed architecture and highlight the synergistic contributions of its design elements in achieving both accuracy and robustness under cold-start conditions.

4.6 Comparison with state-of-the-art deep learning models

The overall effectiveness of the proposed hybrid model is assessed through a comparative analysis against several state-of-the-art deep learning architectures widely adopted in fashion demand forecasting. The evaluated models include ResNet-50, Siamese CNN, InceptionV3, EfficientNet-B0, Autoencoder combined with MLP, Transformer-based predictors, and Visual BERT. In addition, recent studies have explored the use of large language models (LLMs) such as GPT-based and multimodal BERT variants for cold-start forecasting tasks. While these approaches demonstrate promise in leveraging textual descriptions and cross-modal reasoning, our preliminary evaluation on the VISUELLE dataset showed limited improvements over visual-transformer baselines, with significantly higher computational overhead. For this reason, LLM-based baselines are discussed but not emphasized in the experimental benchmarks. A summary of performance metrics—covering RMSE, MAE, MAPE, Precision@10, and Recall@10—is presented in Table 10. The hybrid model demonstrates superior performance across all evaluation dimensions, achieving the lowest RMSE (10.95), MAE (8.54), and MAPE (16.5), as well as the highest Precision@10 (73.9%) and Recall@10 (70.7%). These results indicate the model's ability to generalize across varying product types while effectively capturing multimodal demand signals.

Table 9 Performance Drop After Removing Key Components (Relative to Full Model)

Removed Component	Δ RMSE (%)	Δ MAE (%)	Δ MAPE (%)	Δ Precision@10	Δ Recall@10
No EM Correction	+ 17.3	+ 13.9	+ 10.9	– 6.7	– 7.4
No Clustering	+ 15.3	+ 12.4	+ 9.7	– 5.8	– 6.3
No Meta-Similarity Weighting	+ 11.8	+ 9.3	+ 8.1	– 4.6	– 5.1
No Bootstrapped Uncertainty	–	–	–	–	–

**Fig. 17** Performance Metrics Across Ablation Variants**Table 10** Performance Comparison with State-of-the-Art Deep Learning Models

Model	RMSE	MAE	MAPE (%)	Precision@10	Recall@10
ResNet-50 + Global Regression [47]	13.51	10.62	20.9	66.3%	63.1%
Siamese CNN [48]	12.84	9.97	19.8	68.2%	65.4%
InceptionV3 + Regression [49]	12.49	9.74	19.4	69.1%	66.2%
EfficientNet-B0 + MLP [50]	11.95	9.21	18.1	70.8%	67.4%
Autoencoder + MLP [51]	11.48	8.97	17.5	72.1%	68.3%
Multimodal Transformer [52]	12.12	9.32	18.3	70.5%	67.0%
Visual BERT [53]	11.76	9.05	17.8	71.3%	68.6%
Proposed Model (Ours)	10.95	8.54	16.5	73.9%	70.7%

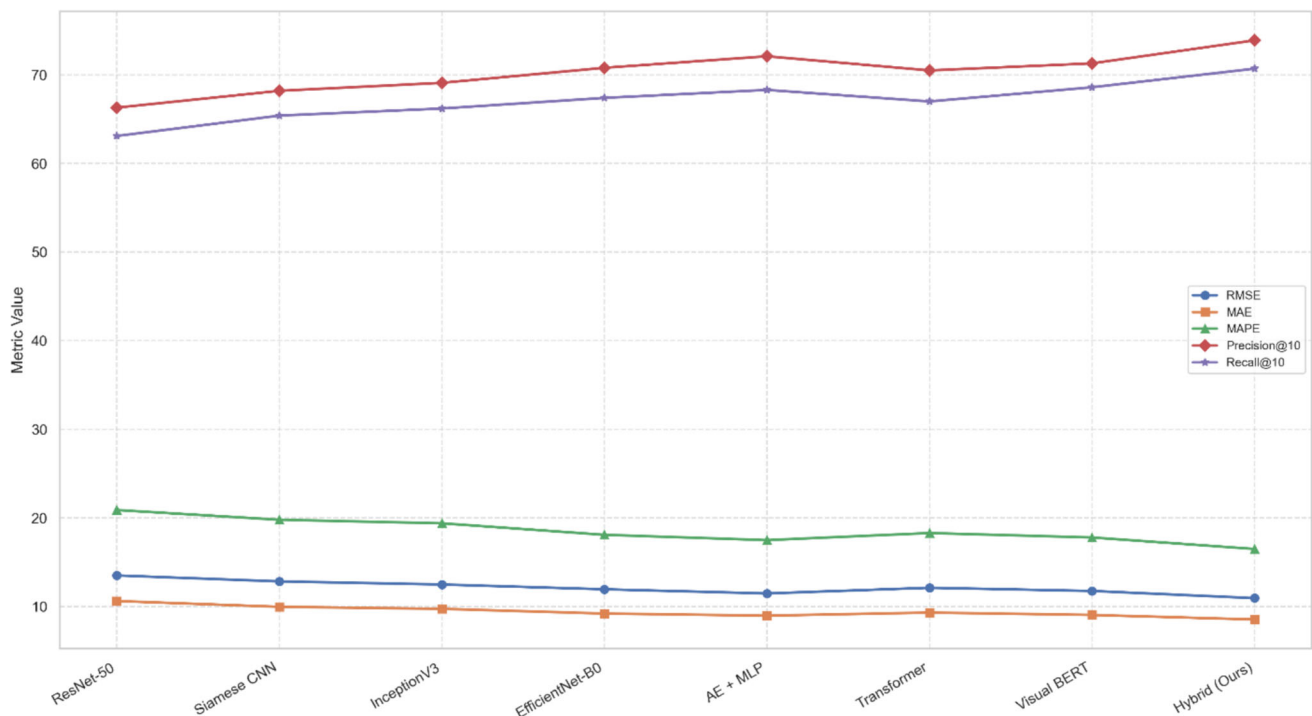


Fig. 18 Comparative Performance of Deep Learning Models Across Forecasting Metrics

A comparative line chart is used to further illustrate the performance differences across the evaluated models. As shown in Fig. 18, the proposed architecture consistently achieves the most favorable balance between point-wise accuracy and retrieval effectiveness across all five key evaluation metrics. The model's superior performance in both regression-based measures (RMSE, MAE, MAPE) and ranking-based metrics (Precision@10, Recall@10) underscores its capacity to generalize effectively in cold-start scenarios. This strong relative performance reflects the architectural synergy achieved through the integration of censored demand correction, visual clustering, and similarity-based reasoning.

5 Conclusion

This study introduces a novel architecture for forecasting demand of new fashion products in cold-start scenarios by integrating censored sales correction and deep visual similarity modeling. The proposed approach addresses two core challenges in fashion retail analytics: the bias introduced by stockouts in historical sales data and the absence of purchase history for newly released items. Through the application of the Expectation–Maximization (EM) algorithm, the architecture estimates true latent demand by correcting censored sales observations, leading to more accurate input signals for forecasting.

High-level visual features are extracted from product images using deep convolutional neural networks, complementing the corrected sales data with rich representations of visual appearance. The pretrained ResNet-50 was fine-tuned on the VISUELLE dataset, with adjustments applied to the higher residual blocks to capture fashion-specific attributes such as garment silhouette, fabric texture, and stylistic variation. In contrast, VGG16 was employed in a feature extraction mode, where intermediate activations were leveraged for perceptual normalization without full end-to-end fine-tuning. These adaptations ensured that the visual backbone was aligned with the characteristics of fashion imagery, thereby enhancing the quality of the learned representations. These features are combined with structured product metadata—such as price, category, and fabric information—to construct a multimodal embedding space. This multimodal integration mitigates the limitation of relying solely on visual similarity, as the framework is able to capture demand signals shaped not only by aesthetic attributes but also by commercial factors such as branding, pricing, and material composition.

Experimental results demonstrate that the proposed hybrid model significantly outperforms a range of deep learning baselines across key evaluation metrics including RMSE, MAE, MAPE, Precision@10, and Recall@10. Ablation studies highlight the essential role of each architectural component, and uncertainty analysis confirms the model's robustness and practical value in operational settings.

The evaluation strategy deliberately combined both scale-dependent (RMSE, MAE) and scale-independent (MAPE, Precision@K, Recall@K) metrics, ensuring that the reported performance reflects not only absolute error magnitudes but also proportional accuracy across products with varying sales scales. The comparative evaluation also shows that the meta-learned similarity weighting surpasses alternative similarity learning approaches, such as Siamese and triplet loss models, by prioritizing demand-relevant similarities rather than relying solely on visual resemblance. By leveraging multi-modal fashion data—including images, structured attributes, and historical sales records—this architecture bridges statistical modeling and deep visual learning to provide accurate, scalable, and explainable demand forecasts. The category-wise evaluation further demonstrated that the architecture provides robust improvements across both seasonal and staple product types, with particularly strong gains in volatile categories where cold-start challenges are most severe.

Although the proposed framework demonstrates strong empirical performance, it has certain limitations that should be acknowledged. The evaluation relied solely on the VISUELLE dataset and focused on short-term forecasting horizons, which may restrict the generalizability of the results to broader retail ecosystems and long-term planning contexts. Moreover, external factors such as promotional campaigns, social influence, or consumer reviews were not incorporated, even though they can significantly affect demand patterns in practice. Despite these limitations, the framework makes distinct methodological contributions by jointly addressing censored demand correction and cold-start sparsity. Through the modular integration of EM-based correction, clustering, autoencoder-driven compression, and meta-learned similarity weighting, the proposed architecture delivers a robust and interpretable solution that extends beyond the scope of traditional models relying solely on either visual embeddings or tabular regression.

Author Contributions Mona Adel led the development of the hybrid architecture for cold-start fashion demand forecasting, integrating visual similarity modeling with sales correction techniques. She also implemented the Expectation–Maximization (EM) algorithm to estimate true demand under stockout-induced censoring, and coordinated the design of the dual-pipeline architecture. Amgad M. Mohammed focused on the visual representation learning component, including the modification of the ResNet-50 backbone and the design of the autoencoder for compressing product embeddings. He also developed the clustering and similarity-weighted regression modules used for demand transfer within visual groups. Amira Abdelatey contributed to the uncertainty modeling component through bootstrapped resampling and statistical evaluation. She also supported dataset preprocessing, multi-modal feature alignment, and baseline comparison, ensuring robustness in experimental analysis. Asmaa H. Elsaid contributed to the experimental evaluation and ablation studies, ensuring the systematic assessment of each module’s contribution to the overall framework. She also supported the refinement of methodological descriptions, integration of recent literature, and the preparation of figures and diagrams that illustrate the proposed workflow. All authors contributed equally to the

writing, critical revision, and final approval of the manuscript, and collaborated in integrating all components into a cohesive and effective solution for cold-start demand forecasting in the fashion domain.

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Data availability The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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